



City, University of London

MSc in Computer Games Technology

Project Report 2023

**A flexible and modular platform for presentation of medical data to
clinicians**

Shreyas Manjunath - 220040817

Supervised by: Philip Edwards

Submitted on: 10th December 2023

Declaration

By submitting this work, I declare that this work is entirely my own except those parts duly identified and referenced in my submission. It complies with any specified word limits and the requirements and regulations detailed in the assessment instructions and any other relevant programme and module documentation. In submitting this work, I acknowledge that I have read and understood the regulations and code regarding academic misconduct, including that relating to plagiarism, as specified in the Programme Handbook. I also acknowledge that this work will be subject to a variety of checks for academic misconduct.

Signed:

Shreyas Manjunath

ABSTRACT

This project explores the development and deployment of a mixed reality application for medical data visualization using Microsoft HoloLens. HoloLens offers an exciting new way to immerse in mixed reality learning environments. The application serves as a technical platform for students and healthcare professionals to interact with three-dimensional anatomical models and screens. User testing is conducted to compare the effectiveness of two-dimensional versus three-dimensional visualisation components in improving learning and interaction experiences. The project also investigates user interaction implementation, ease of use, and its potential impact on educational outcomes in the medical field, focusing on intuitive design and user engagement.

Keywords:

Mixed reality, Stereoscopy, Anatomical, Visualisation, Software Engineering

ACKNOWLEDGEMENTS

I would like to thank my supervisor Philip Edwards for his assistance in developing and completing this project as well as for his expert guidance throughout; my friends for their support and acceptance during the times I missed out on social events.

My parents, thank you for making all of this possible.

TABLE OF CONTENTS

ABSTRACT	i
ACKNOWLEDGEMENTS	i
LIST OF FIGURES	vii
Table of figures	ix
1 Introduction	1
1.1 Background.....	1
1.2 Motivation.....	2
1.3 Research questions	3
1.4 Objectives	3
1.5 Beneficiaries	3
1.6 Project outcomes	4
1.7 Work plan	4
1.8 Major changes.....	5
1.9 Project structure	5
2 Context	7
2.1 Background.....	7
2.4 Overview of Mixed reality.....	10
2.2 Applications of HoloLens in medical areas.....	11
2.3 Software Engineering	12
2.4 Spiral Model.....	12
2.5 Universal windows platform.....	14
2.6 Technical foundations	14
3 Methods.....	17
3.1 Development Method.....	17
3.2 Development Environment.....	17
3.3 Backups	18
3.4 Workflow for Mixed Reality Application Development	19
3.5 Conceptual overview.....	19
3.6 Project settings.....	22
3.7 Build and deploy settings for HoloLens.....	23
3.8 Implementation	24
3.9 HoloLens gestures.....	33
3.10 User testing	35
4 Results	42
4.1 Platform Images	42
4.2 Diagrams.....	54
4.3 Summary of key results	56
5 Discussion.....	57
5.1 Overall objectives.....	59
5.2 Known issues with HoloLens	59

5.3 Recommendations	59
5.4 Summary of the lessons learnt	60
6 Evaluation, Reflections and Conclusions.....	61
6.1 Project management	61
6.2 Future work.....	61
6.3 Reflection.....	61
6.4 Conclusion	62
REFERENCES	63

Appendix Table of Contents

Appendix A: Project Proposal for MSc in Computer Games Technology	A1
Appendix B: Participant Consent Form	B1
APPENDIX C: Participant Information.....	C1
APPENDIX D: Participant Questionnaire.....	D1
APPENDIX E: Participant C - Questionnaire	E1

LIST OF FIGURES

Figure 1 HoloLens device	10
Figure 2 Virtuality Continuum	10
Figure 3 Spiral development model	12
Figure 4 Spiral model in action.....	13
Figure 5 A sketch of UWP API	14
Figure 6 Workflow components diagram for Mixed Reality Application Development	19
Figure 7 Unity sample scene	20
Figure 8 Unity's Plane GameObject	20
Figure 9 Mesh render for Plane GameObject	21
Figure 10 Box collider.....	21
Figure 11 Unity Layers menu	22
Figure 12 Unity quality settings for HoloLens	22
Figure 13 Target Windows Holographic SDK	23
Figure 14 Unity's project build settings	24
Figure 15 Layers for the cameras.....	25
Figure 16 Target eye for the cameras.....	26
Figure 17 Clear flags set to solid color and Clipping Planes set to 0.85.....	26
Figure 18 plane at points $\langle 0,0,0 \rangle$ in world space source.....	27
Figure 19 code snippet from Bounding.cs.....	28
Figure 20 Manipulation mode	29
Figure 21 Alignment type.....	30
Figure 22 Scene content adjuster script.....	30
Figure 23 Layer for the parent GameObject.....	31
Figure 24 InputManager prefab in unity hierarchy and in the scene	32
Figure 25 DefaultCursor prefab from Holotoolkit	32
Figure 26 Default cursor assigned as pointer interface to Input manager	33
Figure 27 Mixed reality camera prefab	33

Figure 28 Bloom close	34
Figure 29 Bloom close	34
Figure 30 gaze and air-tap gestures	35
Figure 31 Page 1 of user questionnaire	36
Figure 32 Page 2 of user questionnaire	37
Figure 33 Page 3 of questionnaire	38
Figure 34 Questionnaire scores given by the students	39
Figure 35 Bar chart for understanding procedures using 3D and 2D screens	40
Figure 36 bar graph for area of recommendation	41
Figure 37 Unity Scene display with camera preview.....	42
Figure 38 Project Hierarchy	43
Figure 39 Hierarchy extended for KidneyCollection	43
Figure 40 project's assets organisation.....	44
Figure 41 2D screen on the left and 3D screen on the right	48
Figure 42 Anatomical model with textures	48
Figure 43 Cursor pointed on the screen	49
Figure 44 Cursor pinched.....	49
Figure 45 cursor pointed on the 3D screen	50
Figure 46 cursor pinched for interaction.....	50
Figure 47 Cursor pointed on the model.....	51
Figure 48 cursor pinched for interaction.....	51
Figure 49 More pictures of the user testing.....	53
Figure 50 One participant tested the HoloLens while seated.....	53
Figure 51 Gaze gesture system flowchart	54
Figure 52 Sequence diagram for grab and move interaction	55

Tables

Table 1 Research questions of this study 3

Table 2 Unity's Hierarchy..... 43

Table 3 Assets categorisation 44

Table 4 cursor interaction with 2D screen..... 49

Table 5 Cursor interaction with 3D screen..... 50

Table 6 Cursor interaction with 3D model 51

1 Introduction

This is a mixed reality project that serves as the fulfilment of the Individual Project component within the curriculum of the MSc in Computer Games Technology program at City University of London. The aim was to develop a mixed reality application for visualization of medical data to clinicians using HoloLens. HoloLens, a mixed reality device, represents a novel innovation in the medical field for both study and practice. The user is directly presented with a range of visuals and data through the HoloLens. They are being used in the medical field to improve medical professionals' educational and professional experiences.

The main goal of this project is to explore the effects of 3D perception and interactivity in a mixed reality environment, specifically using the Microsoft HoloLens 1 headset, on educational outcomes in a surgical context. The project intends to explore whether 3D rendering of a surgical scene, combined with an anatomical model of the patient, can enhance the education and understanding of the spatial positioning of various anatomical structures.

A key aspect of the study is to determine if the ability to move and align the anatomical model within the mixed reality environment leads to a better understanding of the surgical scene for the participants. This could have implications for the effectiveness of mixed reality as a tool in medical education and training, particularly in complex areas such as surgery where understanding spatial relationships is crucial.

1.1 Background

Both augmented and virtual reality applications come under the category of mixed reality (MR). Real-time interaction and coexistence of digital and physical objects blend the virtual and real worlds in mixed reality. Microsoft's HoloLens, the first mixed reality device on the market, gives surgeons hands-free intraoperative access to complicated data, the actual world, and two-way communication. In the realistic 3D visualization environment, MR technology can offer a perfect platform for use in operating rooms for surgical guidance. With the aid of virtual 2D screens and MR technology, surgeons can view a patient's data that has been collected intraoperatively (during surgery) or preoperatively (before surgery) (Cartucho et al., 2020).

Few of these applications have been effectively incorporated into routine clinical practice, despite their known potential. The ability to overlay virtual objects projected by a dedicated device over a real world is the most defining feature of augmented reality (Bergonzi et al., 2019). However, it is best to arrange the virtual image side by side rather than overlaying it onto the real view, as overlaying the virtual objects confuses the scene (Dilley et al., 2019). It is difficult for physicians to be completely engaged in a situation while using virtual reality (VR), as it is limited to digital visuals and cannot be used in real-world treatments.

The project provides a modular method that facilitates the use and interaction of medical professionals and novices with the visualization components. In order to mentor more junior doctors, the application will involve discussing completed surgical cases. The aim of this project is to educate medical students and users on the appropriate placement of visual components, give them control over it, and let them decide how to use the application.

1.2 Motivation

This project was motivated by the opportunity to work with an innovative technology that helps to provides an improved user experience for clinicians. During the development of this project there was no prior guide for the implementation of a 3D video and only a limited knowledge base by fellow developers. The creation of the suggested project for healthcare education may help to advance medical education. This feeling of direction will be incredibly gratifying and inspiring.

Although not directly related to engineering, developing an immersive virtual learning environment for medical education will stretch my capabilities in user experience design, interactive 3D environments, and intuitive interfaces.

Taking on a mixed reality project will be an interesting challenge for me as a student of computer game technology. It will encourage me to learn new skills, methods, and technologies that will broaden my understanding of unity game engine.

1.3 Research questions

The research question underlying this project are represented in the table below.

Table 1 : Research questions of this study

#	Research Questions (RQ)
RQ1	Does 3D visualisation of the video help in understanding the procedure?
RQ2	Does 3D aligning of the patient anatomy with a surgical scene help in understanding the procedure better?
RQ3	Does giving the ability to interact with medical data be helpful in medical setting?
RQ4	How can we involve this visualization platform in the healthcare setting to provide improved learning over traditional approaches?
RQ5	How can we use visualization components such as 3D anatomical model and 2D/3D screens to benefit clinicians and novices?

1.4 Objectives

The primary objectives of our project involve multiple activities, we are looking into and implementing medical visualization tools within the Unity platform, as well as conducting research on online clinical datasets. In addition, we are allowing visual elements to be viewed simultaneously on 2D screens and 3D screens with 3D virtual objects. Furthermore, we want to involve both users and medical professionals in order to learn about the advantages that they see in each visualization technique. This approach guarantees a comprehensive and useful application of visualization tools in a medical setting, improving our developed application's usefulness and user experience.

1.5 Beneficiaries

Medical professionals, including surgeons, physicians, nurses, and medical students, are the main beneficiaries of the research findings. It is hoped that this research will offer valuable insights not only to these healthcare professionals but also to the collaborative

efforts of clinicians and engineers engaged in the development of innovative mixed reality projects. The aim is for this review to provide a form of assistance to individuals working on novel mixed reality initiatives in the medical field.

1.6 Project outcomes

The primary outcome of this study is the 3D stereo video implementation and the user testing of it. This study conducts a comparative analysis of 3D anatomical models against 2D and 3D screens, focusing on enhancements in visualization and interaction. Through implementing and user-testing 3D stereo videos, it evaluates the efficacy and limits of 3D technology in real-world applications, thus offering key insights into its practical advantages and constraints.

1.7 Work plan

The project development faced numerous technical challenges and worked with the HoloLens 1 device in this project. The complete work plan is available in Appendix A, the last section of the proposal document, it contains the entire Gantt chart, it illustrates the project's flow from creation to completion by providing a timeline and duration for each task as well as how the phases overlap. Major phases of this project are explained below:

- **Phase 1: Research and Planning:** The project commenced with an extensive literature review, studying existing works related to various game engines and the offerings of mixed reality toolkit packages. Evaluation was done on Unity versions to determine the compatibility with the mixed reality toolkit. A decision was made to prioritize the 3D stereo visualization component and implementation of MRI/CT slice views was set for future development. Due to the unavailability of HoloLens 2 and the author's system incompatibility with the HoloLens emulator, we chose to use HoloLens 1.
- **Phase 2: Platform Development:** once the Unity engine and Holotoolkit package versions were defined, we set up the platform for HoloLens development and deployment. The platform is tested for deployment with simple projects and successfully deployed a cube onto HoloLens 1 following several attempts.

- Phase 3: Phase 3: Testing and Evaluation: To ensure the platform's usability and functionality, the project is put through a rigorous testing phase during this phase. In order to refine and enhance the project, this entails conducting user testing and obtaining input.
- Phase 4: Finalization and Deployment: we will prepare the platform for deployment, make final adjustments based on received feedback, prepare for deployment, and submit the project for assessment. We conclude the project by preparing the project report and documentation.

1.8 Major changes

The project successfully implemented a mixed reality application with 2D and 3D screens in addition to an anatomy model. However, it is important to note a major change from the original plan: for the current project phase, the decision to delay the MRI/CT scan slice view feature's implementation was made as the project moved forward, planning to include it in future development efforts. This decision was primarily influenced by existing time constraints, acknowledging the complexity and time required for successfully integrating a sophisticated feature. This approach allows for a more focused and manageable current development phase, while setting a clear roadmap for future enhancements.

1.9 Project structure

Chapter 1 - Introduction: provides a brief introduction of the project and its background. It also defines the problems it hopes to answer, along with the expected outcomes and changes to the project.

Chapter 2 – Context: examines the project's history and highlights the important works that have been written about it, showing how those works influenced the decisions made during the project.

Chapter 3 – Methods: The development process is explained along with the complete application development, a description of the implemented visual components with their

user interaction and an explanation of the rendering techniques employed. Finally, a discussion of the user evaluation approach is addressed.

Chapter 4 – Results: The project's outcomes are described in this chapter, with special attention to the mixed reality application in use. It includes important diagrams, explanations, and pictures of some of the main results.

Chapter 5 – Discussion: compares the initial goals with the completed application and describes the degree of achievement of each goal. Where the final project deviated from the project plan, specifics are given, along with some important takeaways and suggestions.

Chapter 6 –Evaluation, Reflections and Conclusions: The entire project is evaluated, conclusions are drawn, and the knowledge acquired from creating the game is summed up. Lastly, a few suggestions for additional research are given.

In addition to these chapters, Appendix A contains a copy of the initial project proposal, Appendix B contains the participant consent form, and Appendix C contains participant information. Additionally, the questionnaire is provided as the final appendix D.

2 Context

This chapter provides a broader context for the project. It begins with an overview of mixed reality, followed by a focused review of its uses in the medical field. Subsequently, previous studies in medical visualization and augmented reality are discussed, emphasizing the significance of visualization applications in medical training. The chapter delves into existing challenges related to data quality, and 3D model accuracy. Additionally, it reviews the role of medical image processing and 3D visualization in addressing these challenges.

2.1 Background

In the last few years, augmented reality (AR), virtual reality (VR) and mixed reality (MR) have become major technologies, and these technologies provide immerse experience by producing realistic sounds, images, and sensations. The ability to overlay virtual objects projected by a dedicated device over a real world is the most defining feature of augmented reality (Bergonzi et al., 2019). However, it is best to arrange the virtual image side by side rather than overlaying it onto the real view, as overlaying the virtual objects confuses the scene (Dilley et al., 2019). It is difficult for physicians to be completely engaged in a situation while using virtual reality (VR), as it is limited to digital visuals and cannot be used in real-world treatments. If mixed reality (MR) can accurately deliver patient-specific data to clinicians to help guide the procedure, it could greatly benefit many clinical procedures, on the other hand, if the visual content is superimposed incorrectly, it may mislead the physician and possibly cause a catastrophic medical incident (Chen et al., 2017).

Mixed Reality (MR) is becoming more widely used in a variety of medical applications. Microsoft created the state-of-the-art mixed reality (MR) device known as the HoloLens in 2016. It is an MR-based head-mounted display (HMD). With the use of holograms, this wearable gadget allows users to engage with their surroundings while thoroughly engaging all of their senses. More sophisticated features of the HoloLens include gaze design, gesture recognition, spatial sound creation, spatial mapping, and stereoscopic

three-dimensional (3D) projection hence, multiple studies have been carried out to find more effective ways to use the HoloLens based on these features (Park et al., 2021).

This study isn't the first instance where virtual 3D models have been utilised to assist in medical procedures. For example, multiple research groups have previously introduced such as, (Kumar et al., 2020) focused on developing patient-specific models by segmenting CT/MRI scans to create 3D surface models of organs utilizing HoloLens for medical data visualisation. Surgeons found HoloLens beneficial, and it was used in the operating room to visualize 3D anatomical models during medical operations and surgery preparation. By using 3D holograms for preoperative planning during brain tumour surgery, the study by (Incekara et al., 2018) sought to assess the clinical viability and accuracy of the HoloLens. It concluded that the HoloLens was clinically practical for usage in the operating theatre. In the research by (Cartucho et al., 2020), surgeons prefer the 3D model as the preferred visualization component. These studies demonstrate how surgical guidance can be assisted by 3D models in visualization applications.

According to (Hyde et al., 2019) the use of interactive virtual 3D models might affect surgical decisions, possibly changing the surgical approach and increasing surgeon confidence. (Nguyen et al. 2020) used HoloLens as an AR technology to measure the accuracy of virtual object placement. They assessed the accuracy using three distinct techniques: keyboard control, 3-point correspondence matching, and tap to place. According to their study, the tap-to-place approach proved to be the most effective. This method depends on the user's hand gesture accuracy to select a point in a 3D world, which can be difficult in complex or complicated surroundings. This could result in the placement of virtual items being less precise. However, this project's grab-and-move interaction closely mimics movements in the real world, allowing users to interact with virtual objects naturally and directly. The user feels more in control and immersed in the virtual environment when using this technique. (Jannin & Collins, 2013) provided an excellent review of mixed reality in image-guided surgery, who also stressed the significance of employing the right visualization processing methods.

The utilisation of 3D visualization in medical training and procedures significantly enhances depth and clarity for learners and practitioners, thereby improving medical education and procedural skills. (Yammine & Violato, 2014) meta-survey revealed that,

in terms of factual and spatial knowledge, 3D visualisation techniques are superior to traditional methods when it comes to learning and teaching anatomy. Despite this advancement, studies comparing the effectiveness of various 3D visualisation techniques are lacking. In order to address this, the project offers a platform on which a 3D model and 3D stereo video visual elements are applied in a setting and a questionnaire is developed. This survey asks for opinions on the usefulness and possible enhancements of particular visualizations. In addition to assessing the efficacy of 3D visualisation techniques as they are today, this method solicits ideas for future improvements with the goal of adapting these instruments to the changing requirements and preferences of users.

2.1.1 Mixed reality device

Microsoft (Redmond, Wash.) introduced HoloLens™ (Figure 1), the first mixed reality headset that was made available to the general public, in March 2016 (Tepper et al., 2017). The HoloLens stands as a leading mixed reality head-mounted device in the market. A number of sensors are built inside the HoloLens, such as an ambient light sensor, a time-of-flight (ToF) depth sensor, four microphones, an inertial measurement unit (IMU), four visible light cameras oriented sideways to record the environment, and a front-facing, high-definition photo/video camera (Gsaxner et al., 2023). To give the user three-dimensional vision, stereoscopic virtual content is projected over the real world in two combiner lenses that are semi-transparent. For the best user comfort and image quality, it is advised to position virtual items near to the user's 2 meter optical focus distance from the HoloLens displays (Gsaxner et al., 2023). Built-in speakers and a 32-bit Intel Atom x5 CPU with 1 GB of random access memory provide audio and is associated with 64 GB of storage on the HoloLens and two to three hours is the estimated active battery life (Gsaxner et al., 2023). By tracking the users' head movements, the direction of their gaze can be determined. Voice commands and hand gestures are two ways that users can interact with virtual objects that are automatically recognised. The HoloLens was first promoted for use in 3D modelling, gaming, and communication; nevertheless, the medical field soon became interested in it.



Figure 1 HoloLens device (Source: Lolambean, 2021)

2.4 Overview of Mixed reality

The phrase "mixed reality" was first used in Paul Milgram and Fumio Kishino's 1994 book "A Taxonomy of Mixed Reality Visual Displays" (qianw211, 2023). The following diagram is produced from Draw.io (Draw.io) by referencing (Milgram et al., 1995).

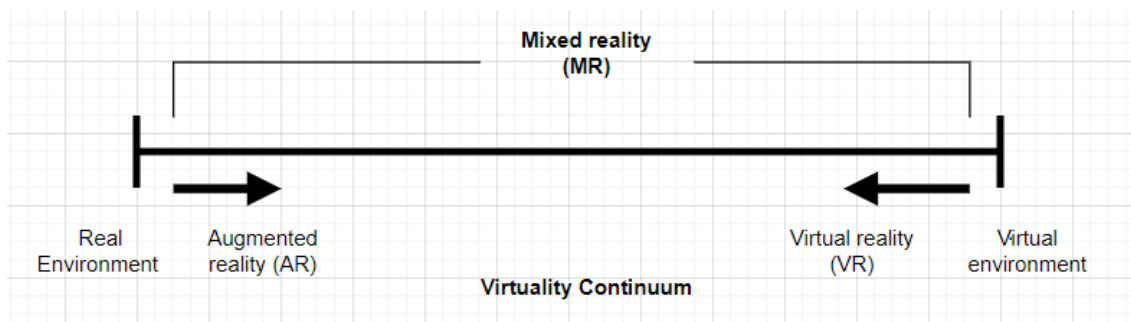


Figure 2 Virtuality Continuum

The virtuality continuum is a spectrum that has these two realities at its ends. Real environments, like the one in which humans live, are at one end of the spectrum, and at the other is the corresponding virtual reality. In this context, it is simple to characterize. Anywhere in between the extremes of the virtuality continuum, or on a single display, objects from the real and virtual worlds are displayed together, making an environment that is generically known as Mixed Reality (MR). (Milgram et al., 1995).

Windows Mixed Reality experiences can be delivered on two main types of devices:

1. A holographic device, a technology that produces or shows holograms, which are three-dimensional images formed when light beams collide. The ability of

holographic devices to display digital objects as though they were real-world objects characterizes them. Holographic technologies give consumers realistic and engaging visual experiences and are widely used in a variety of applications, from industrial design to medical imaging to entertainment and gaming. Microsoft's HoloLens and Magic Leap One are two examples of holographic technology.

2. Immersive VR technologies aim to immerse users in a virtual environment where they may explore, engage with, and interact with objects as if they were real. The ability of immersive virtual reality (VR) devices to obscure the outside world and replace it with a completely immersive digital experience is what defines them as immersive VR devices. Examples of VR device is Meta Quest formerly known as Meta/Oculus Rift, HTC Vive Pro.

2.2 Review of HoloLens in medical areas

Several industries have made use of Microsoft HoloLens since its release in 2016. Practice training, medical and patient education, surgery, rehabilitation, and patient treatment are the top five medical application areas (Chen et al., 2017). AR technology is helpful for teaching students majoring in medical science because it is widely used in medical training and simulation. Despite not being a brand-new technology, because of Microsoft HoloLens, augmented reality (AR) is becoming more common in medical education and simulation (Herron, 2016). Before surgery, patients have benefited from the HoloLens' use in patient education by receiving a more understandable and plausible explanation (Wake et al., 2019, House et al., 2020, Rositi et al., 2020). Similarly, by incorporating 3D models and 3D stereo video visual components, this platform actively aims to perhaps establish a foundation for post-surgery education. These studies have used HoloLens, showcasing the device's potential as an instructional aid. This is consistent with the mixed reality (MR) potential of HoloLens in education, where its ability to combine 3D models with stereo video visualization elements has the potential to transform teaching approaches and provide engaging, interactive learning environments.

2.3 Software Engineering

Software engineering uses proven practices and true methods to formalize and improve the software development process. A software process model's main goal, according to research by (Boehm, 1988), is to specify the order in which the stages of software development must be completed. This study highlights the significance of software process models, mainly because they define the sequence in which a project should carry out its main tasks, such as phases, increments, prototypes, validation tasks, and more. Building software is most effective using an iterative approach like the spiral development model, which facilitates regular assessment and adaptability to unexpected challenges. In this development the spiral model will be applied to make sure that development concentrates on reaching well-defined and known targets.

2.4 Spiral Model

The central development stages are continuously iterated over by the spiral development model, as seen in the figure 2. During these stages, risk analysis is a key component in deciding on the project's process model.

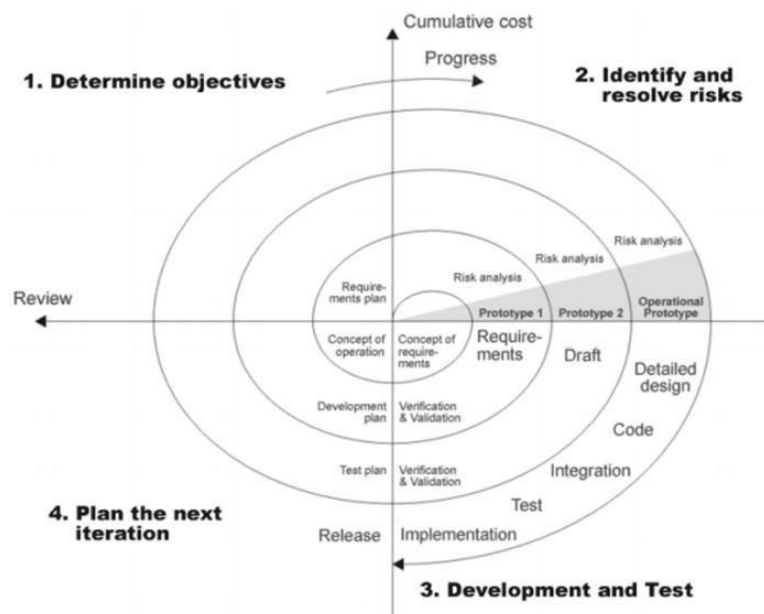


Figure 3 Spiral development model (Source: Young, 2014)

The spiral model in action for this project development can be seen below;

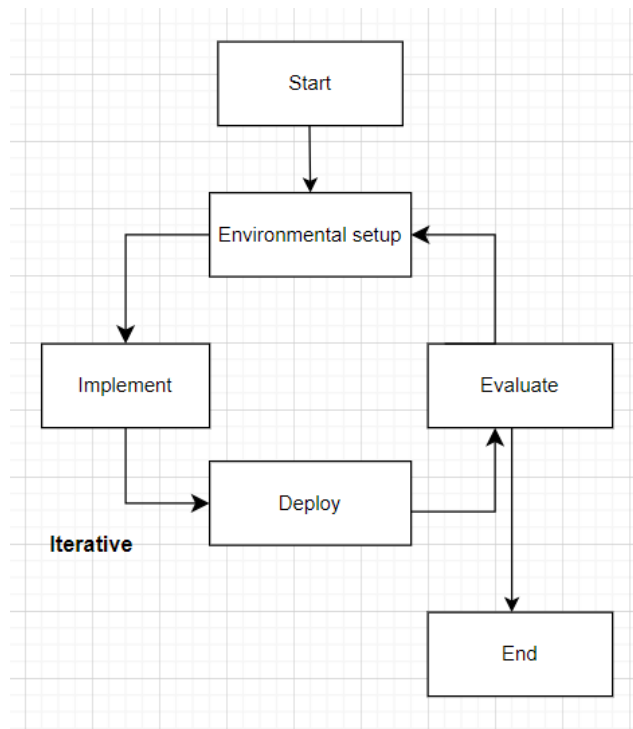


Figure 4 Spiral model in action (produced from Draw.io)

This model can incorporate interactivity and flexibility in this project's development process. The environmental setup is the initial stage in the spiral model (Figure 3) cycle it sets up the environment for the development. Once the environment is setup the features to implement is done in the implement stage. The implemented features are tested and deployed during the deploy phase. Reflecting on what was done right and wrong in the past cycle results in a new set of observations, which then return to the initial step in the evaluate phase.

2.5 Universal windows platform

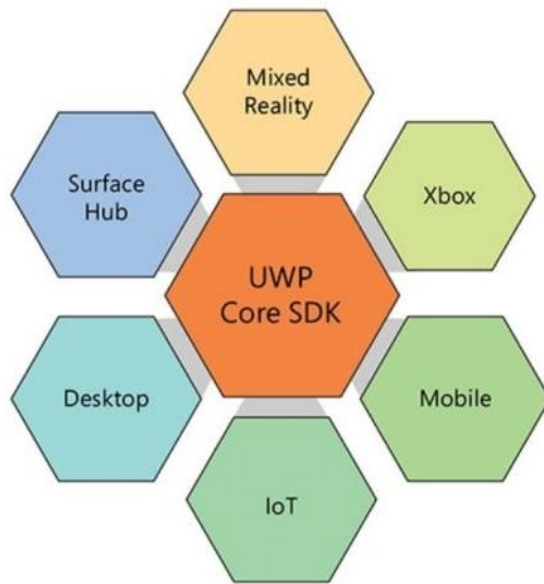


Figure 5 A sketch of UWP API Source: (Borycki, 2019)

Microsoft introduced the Universal Windows Platform (UWP) with Windows 10, which provides a common API to develop applications for a range of device families, including desktop, mobile, Xbox, Mixed Reality, Internet of Things (IoT), and Surface Hub (Borycki, 2019) (Figure 4). Your application can run on any Windows device, be it a desktop computer, Xbox, or mixed reality headset, as long as it only uses the core APIs. An application becomes compatible with all devices running Windows 10 or later when you target the universal APIs. With extension SDKs, you can make calls to device-specific specialised APIs (QuinnRadich, 2023). To target features unique to IoT devices, for instance, you can add the IoT extension SDK to your project if your UWP app is intended for an IoT device. Similarly, this project gains access to specialized tools and APIs by integrating the HoloToolkit extension, which improves its functionality for mixed reality platforms.

2.6 Technical foundations

At the Worldwide Apple Inc. Conference in June 2005, the cross-platform game engine Unity Technologies was unveiled as a Mac OS X-only title (Hussain, 2020). All development scripts are written in C#, while the underlying runtime code is written in C++. Unity 3D is a complete integrated development environment (IDE) that enables the

assembly of art and assets, also known as GameObjects, into three-dimensional scenes. Each GameObject can then have components added to it to give it extra features like animation, physics, lighting, audio, and special effects.

The Holotoolkit or the mixed reality toolkit (newer version) is a crucial component of HoloLens development and is the simplest way to develop in Unity. The Holotoolkit gives developers all the resources they need to begin creating Mixed Reality applications. It accomplishes everything, including setting up Unity's settings and turning on gestures and spatial mapping (Ong, 2017). The study by (Cartucho et al., 2020) made use of packages from the holotoolkit, which is an older version of the Mixed Reality Toolkit. Their study also used Unity 2018.4.7f with Visual Studio 2017 as the programming environment and holotoolkit 2017.4.2 has been used with the HoloLens (1st Gen). The Mixed Reality Toolkit was utilized by (Nguyen et al., 2020) and (Sánchez-Margallo et al., 2021) to improve user interactions in their respective research. While (Sánchez-Margallo et al., 2021) investigated a variety of interactions, including scaling and rotating objects, (Nguyen et al., 2020) concentrated on "tap to place" interactions. The wide range of packages and APIs in the toolkit, which makes it easier to script and implement these interactions, was noted by (Maasthi et al., 2020).

Adding to this, (Turini et al., 2018) created a mixed reality application with the MixedRealityToolkit (version 2017.1.2) that enabled voice commands to be used to interact with virtual contents, HoloLens gestures like Air Tap and Bloom, and head movements like gaze. In a similar way, (Galati et al., 2020) used the MRTK foundation package as the project's core system. As a fundamental framework for HoloLens development in Unity, these varied applications highlight the importance of the Mixed Reality Toolkit and the Holotoolkit, showcasing its adaptability and simplicity in producing immersive mixed reality experiences. This development is further demonstrated by (Bai et al., 2020) who explore various aspects of user interaction in mixed reality settings specifically highlighting the effectiveness of combined gaze and gesture cues over gaze alone. The results of the study demonstrated that in remote collaboration, visual cues have a positive impact on mental workload, performance, social presence, and spatial presence.

Stereoscopy works by providing, two two-dimensional images of the same scene. The left and right eyes receive separate presentations of these images, with minor variations between them. The brain creates the virtual 3D scene by combining these two images to create a sense of depth in a 2D plane. In order to explain this, there is an excellent example by (Borycki, 2019), when you raise your thumb a few inches in front of your face, then rapidly switch between blinking your left and right eyes to get the best sense of this effect. Observe how your thumb appears to be in a different place depending on which eye you use to view it, and how it returns to its proper location when both of your eyes are open. It is the mechanism of stereoscopy. The research by (Takagi et al., 2000) demonstrated that with HMD, it is possible to use a separate screen for each eye. This means that with a stereoscopic display type where both eyes view the same screen, it is possible to set up the display system so that the left and right optical axes intersect.

3 Methods

The entire process of developing an application is covered in detail in the development process, including descriptions of the visual elements and how users interact with them. It also describes the rendering methods that were applied. There is also a discussion of the user evaluation method.

3.1 Development Method

The spiral model (figure 3) is implemented in this project in phases, each of which adds to the finished software. The technical aspects of the mixed reality software being developed, and the user evaluation of the delivered software are the two main areas of evaluation in this implementation and build project.

3.2 Development Environment

At the start of the project, a decision had to be made on which engine to use. The decision would be influenced by factors such as compatibility with targeted hardware, ease of use, available resources, and the engine's community and technical support. Currently there are a number of ways to develop a mixed reality software. The first step is finding the technology that is right. As of 2nd December 2023, Unity and Unreal engines are the two most popular platforms for mixed reality development.

- Unity in C#: one of the most popular MR development platforms, provides a comprehensive set of tools for creating MR experiences. It works with various types of MR hardware, including HoloLens and VR headsets.
- Unreal engine: Unreal Engine 4 is a robust, open-source creation engine that supports mixed reality in both C++ and Blueprints. HoloLens support at present is fully featured and production-ready in Unreal Engine 4.25.
- OpenXR: is a royalty-free, open-source API standard from Khronos that gives engines native access to a large selection of mixed reality device vendors.

Unity was chosen as the preferred engine for this project on the basis of this author's experience with it and the availability of adequate technical support. Although at present Unity 2021.3 LTS is the recommended version for the development of mixed reality application choosing the right version depended upon factors such as project

requirements, stability, feature set, and compatibility. Unity 2018.4 versions work with specific versions of the Holotoolkit, an essential resource for creating mixed reality applications. To ensure optimal functionality, ensure that the Unity version matches the compatible Holotoolkit version.

While the Holotoolkit served as the initial development toolkit for the HoloLens, the Mixed Reality Toolkit is a more advanced and adaptable alternative that provides increased functionality and support for a wider variety of mixed reality platforms. Essential features like gaze and spatial mapping are provided by Holotoolkit and are crucial for the implementation of stereo video and 3D models.

Considering the factors mentioned above and the availability of HoloLens 1st generation as the mixed reality device, unity 2018.4.7f1 LTS was selected as the development engine along with Holotoolkit 2017.4.2.0. This Unity version is compatible with Visual Studio Community 2017 version 15.9.58, which includes the windows 10 SDK (10.0.17763.0) and workloads for C++ desktop development, .NET desktop development, and universal Windows platform development with USB connectivity. Visual studio must also have game development with unity included with visual studio tools for unity, C# and visual basic.

3.3 Backups

To keep track of changes made to the source code and other file collections over time, I decided to use the traditional version control system, Git. I was able to keep a complete copy of the project on my local computer, complete with all of its history, by using Git. I also kept the project files in a private GitHub repository, to which only specific users have access. Additionally, I routinely made thorough backups of the entire directory, including all associated documentation, especially after finishing significant phases.

Crashing occurs when using Unity in the build process. Solving such problems can be very tough because it's frequently hard to determine if the problem is a bug in Unity itself, an error in your own code, or an error in the Unity interface. Consequently, keeping accurate and working copies of the whole project can be reassuring as well as helpful.

3.4 Workflow for Mixed Reality Application Development

Figure 5 shows the workflow components diagram of this mixed reality application in HoloLens. The Unity game engine is the primary development platform utilized in this workflow. It allowed the development of 3D video visualization and user interaction possible. Imported assets include videos and 3D models. The Holotoolkit, a collection of tools, scripts, and components intended to speed up the creation of MR applications, can also be imported into Unity as a custom package. C# scripting is used by Unity with Visual Studio. Unity comes with an integrated Application Programming Interface (API) for mixed reality features like gaze, gestures, and motion controllers. The Visual Studio solution from Unity is compiled to create the application, which is then built and deployed to the HoloLens.

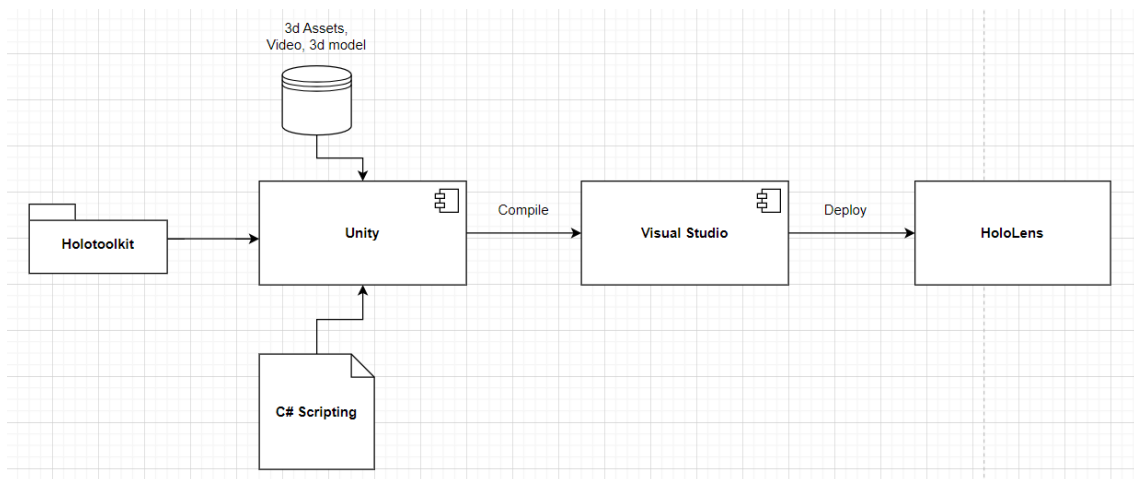


Figure 6 Workflow components diagram for Mixed Reality Application Development (produced from: Draw.io)

3.5 Conceptual overview

The following sections highlight the fundamental ideas and understanding of the concepts and techniques used to develop this application.

3.5.1 Scene understanding

Scenes are where you work with content in Unity. Scenes are assets that comprise all or a portion of an application or game; in Unity, this is where content is worked with (Technologies, 2023). When you create a new project and open it for the first time in Unity, a sample scene with just a camera and directional light appears (Figure 5).

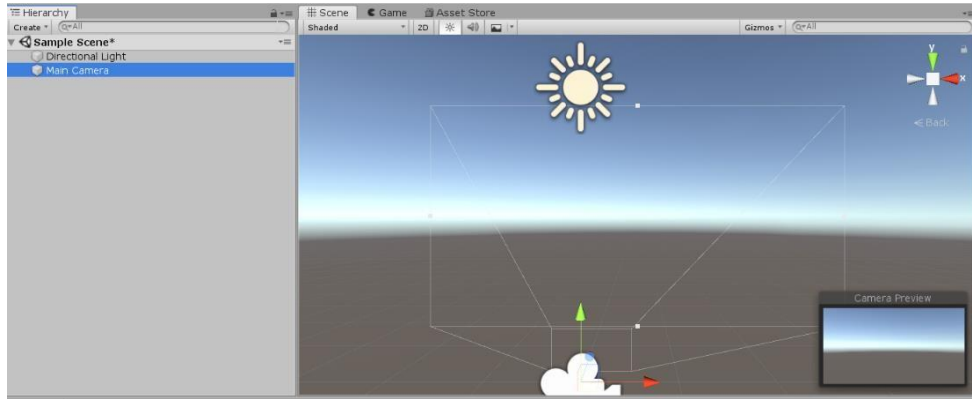


Figure 7 Unity sample scene

3.5.2 GameObject

The most important concept in the Unity Editor is the GameObject. Characters, properties, and scenery are examples of basic objects that are represented by GameObjects in Unity. They don't do much on their own, but they act as holding spaces for Components, which carry out the functionality (Technologies, 2023). The game-object, shown in (Figure 6), is the fundamental unit of the Unity gaming platform. It is a 3D coordinate upon which visualizations can be constructed. The project uses Planes as a 3D GameObject to display stereo video visualization component.

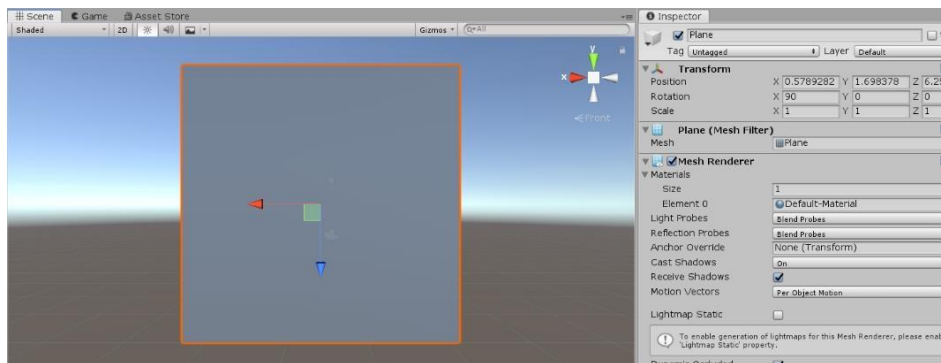


Figure 8 Unity's Plane GameObject

3.5.3 Mesh Renderer

Mesheres are stored by Unity as mesh assets in your project, and they are compatible with various parts. Unity uses Mesh Renderer components to render regular meshes for graphics, and it uses the Mesh Collider component for physics to determine a collider's shape (Technologies, 2023).

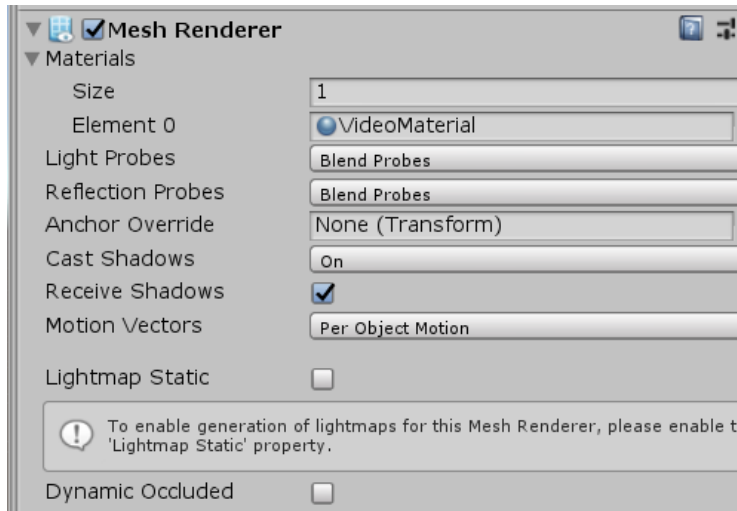


Figure 9 Mesh render for Plane GameObject

A material can also contain information such as color references or texture references if it references a Shader object that defines the material's properties. (Figure 7) shows a custom-made 'VideoMaterial' material added to the plane GameObject.

3.5.4 Box Collider

Rectangular cuboids called box colliders work well with objects like chests and crates. Colliders are used in Unity to specify an object's physical form to facilitate physical interactions and collision detection. A collider such as a Box Collider cannot work or interact with the game world unless it is attached to a GameObject.

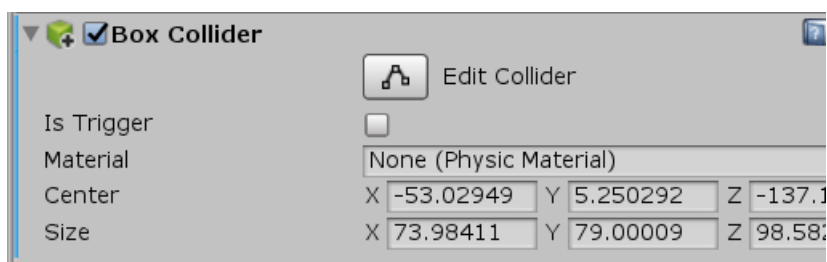


Figure 10 Box collider

3.5.5 Layers

One useful tool for separating GameObjects in your scenes is Layers. Using the Camera's culling mask allows you to render objects exclusively within a specific layer or a selected group of layers in your project. To modify the culling mask of

a selected camera, select it and then select the Culling Mask dropdown in the Inspector window. A layer can be made to stop rendering in the scene by clearing its checkbox. The project generates 3D stereo video by utilizing layers and the culling mask features of the camera, allowing cameras to render each GameObject in a scene. Unity outlines its seven built-in layers, the project makes use of four user layers which is from layer 9 to layer 10 where layer 9 is configured for the left plane and camera, while Layer 10 is configured for the right plane and camera while layer 11 supports the ‘Secondcam’, which shows every gameobjects of the scene (Figure 10).

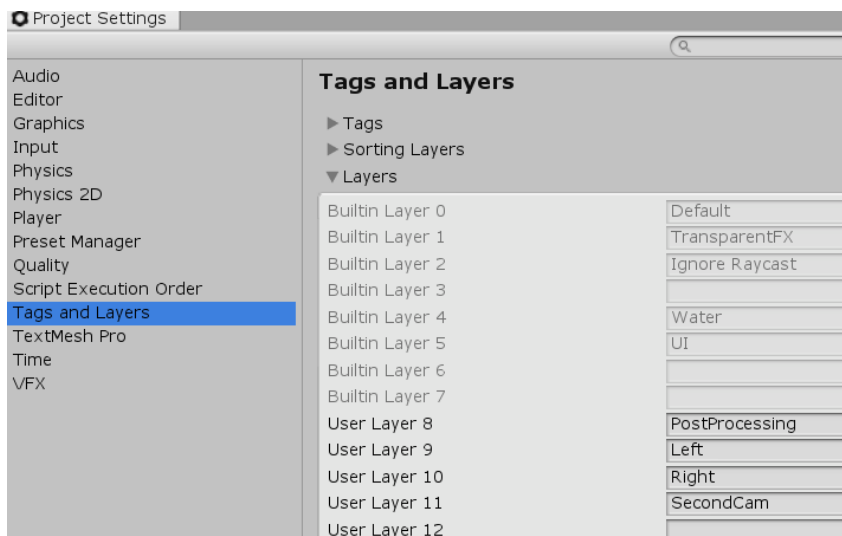


Figure 11 Unity Layers menu

3.6 Project settings

We will use Unity project settings to facilitate the development of a mixed reality platform. We'll also configure our application with a few quality settings. Finally, we will make sure that our build targets are set to the Universal Windows Platform.

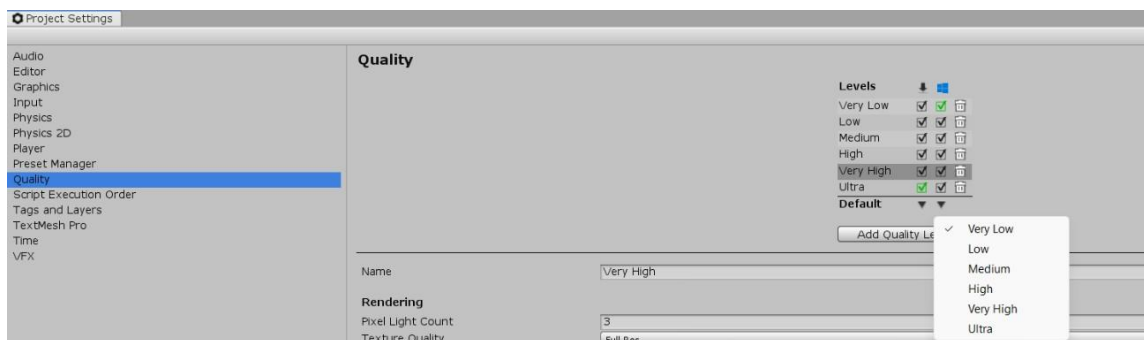


Figure 12 Unity quality settings for HoloLens

On HoloLens, keeping a high framerate is essential, so choose 'Edit > Project Settings > Quality'. Then, select 'Very Low' from the dropdown menu under the Universal Windows Platform logo because we want the quality settings adjusted for optimal performance.

The application we are trying to export should create an immersive view rather than a 2D view, so we are able to configure Unity to support virtual reality when we enable it for the Windows 10 SDK.

Open 'Edit > Project Settings > Player'. Choose the Universal Windows Platform icon from the Player Settings Inspector Panel. Next, expand the XR Settings group. To create a new list of Virtual Reality SDKs, select the 'Virtual Reality Supported' checkbox in the Rendering section. Make sure Windows Mixed Reality is listed; if it isn't, select Windows Mixed Reality by clicking the + button at the bottom of the list.

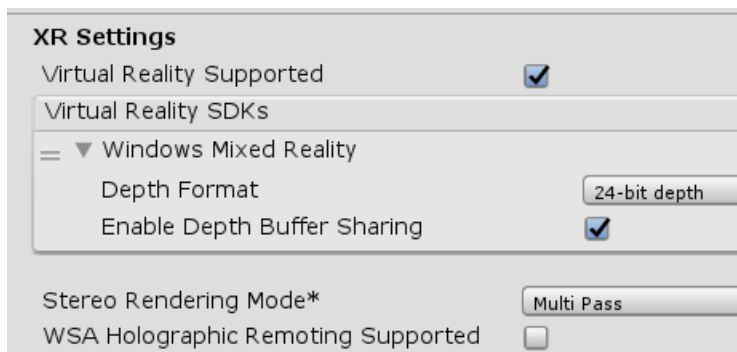


Figure 13 Target Windows Holographic SDK

3.7 Build and deploy settings for HoloLens

The build settings must be selected from the file menu, the scene must be added in the build settings windows, and unity will build the project upon clicking the build button. For the build to be successful, the project platform needs to be on the Universal Windows Platform and the project settings needs to be configured for HoloLens development.

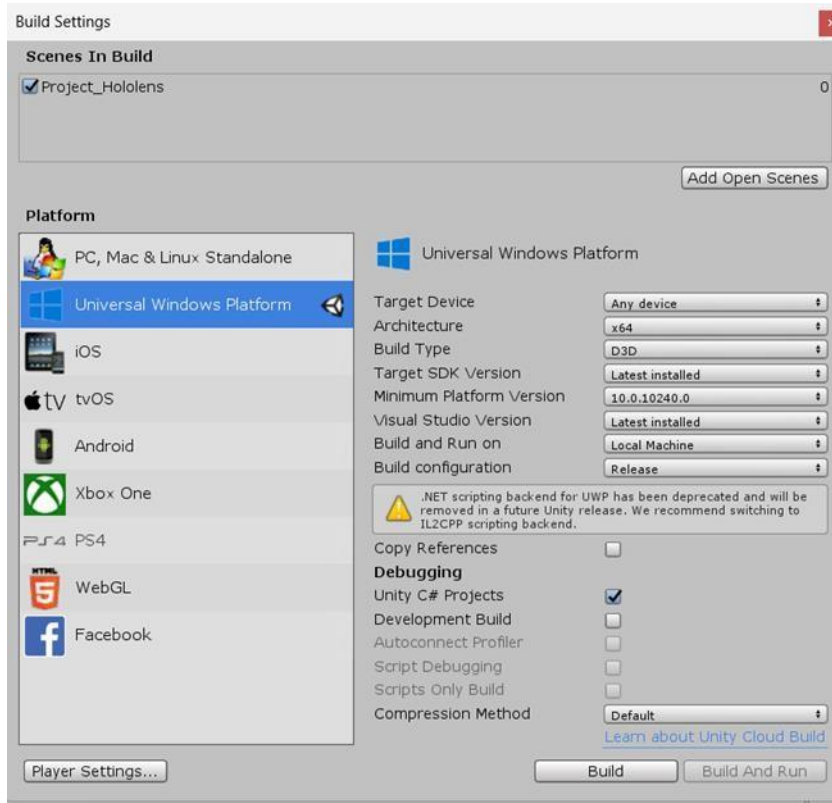


Figure 14 Unity's project build settings

Once the build is completed in Unity, the file explorer will open which will contain a visual studio solution, this solution needs to be deployed from visual studio to HoloLens. There are two ways to deploy not HoloLens.

1. Deploy to HoloLens via USB: The deployment target needs to be set to "Device" with "Release" and "x86" as the target specifications with the HoloLens connected over USB.
2. Deploy to HoloLens via Wi-Fi: the IP address of the HoloLens must be entered with the authentication mode set to universal, and the deployment target must be set to a remote machine.

3.8 Implementation

After installing the required tools (as listed in the development environment), such as the Windows software development kit (SDK), Visual Studio, and the Universal Windows Platform (UWP) module, one can use Unity to develop for HoloLens. Create a new Unity

project and select UWP as the platform after these are installed. Next, modify the project settings to set up development for HoloLens.

3.8.1 3D video visualization component

The visualization of 3D video content was the main focus of the project's first phase of execution. This author dedicated a significant amount of time to this task because it was novel for the research and allowed it to progress freely so that later stages of the implementation could continue.

3.8.1.1 Camera setup

Your holographic world revolves around the mixed reality headset you wear, your head's movement and rotation will be tracked by the unity camera component, which will manage stereoscopic rendering automatically (Keveleigh, 2021). (Unity technologies) mentioned that a camera can be configured to only render specific areas of the screen, in any order, and at any location on the screen. This project also proves that the camera can also be rendered to a specific eye.

As per Unity's documentation, the project's 3D video implementation uses two main cameras, each designated to the 'left' and 'right' layers for selective rendering through the culling mask of the Camera component (Figure 5). To enable stereoscopic 3D visualization, the culling mask and "Target Eye" of the "right" camera are adjusted for the right eye view, and the "left" camera is similarly configured (Figure 6). The two main cameras have the same common settings, with background RGB values set to 0,0,0, clear flags set to solid color Clipping planes set to 0.85 and (Figure 7).

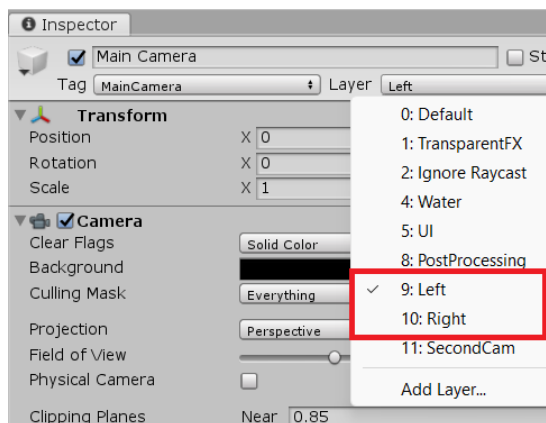


Figure 15 Layers for the cameras

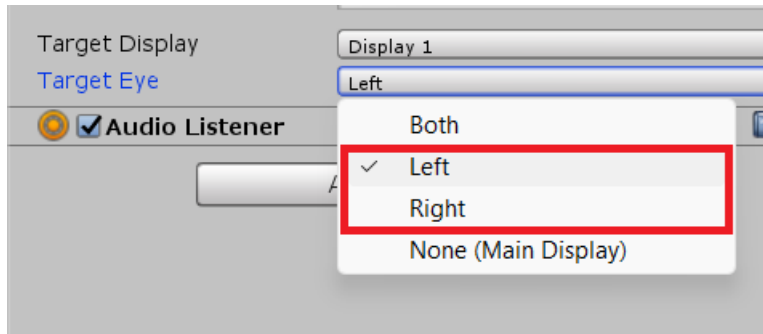


Figure 16 Target eye for the cameras

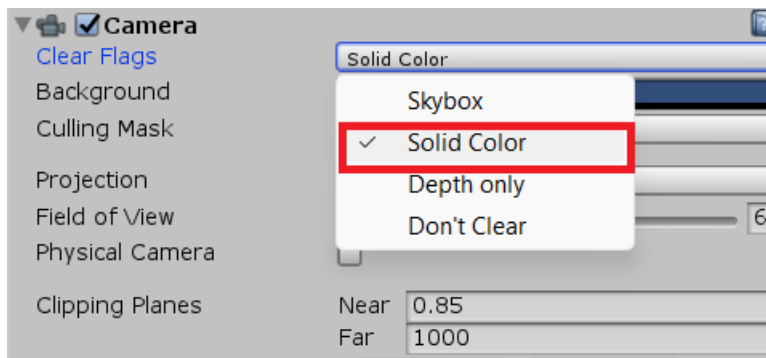


Figure 17 Clear flags set to solid color and Clipping Planes set to 0.85

3.8.1.2 Plane

Similar to the implementation of the two main cameras, there are two planes which plays the same video for each eye. The implementation started by playing videos on each plane using (Nischita, 2019) tutorial, then these planes were tagged with the corresponding layers on the left and right. The project also used (Kunz, 2019) video render texturing guidelines, which helped me texture the planes with the video content in an efficient manner. Once the textures are applied it is mapped onto the 3D plane.

A plane (Figure 7) is defined by a normal vector that specifies its position when it passes through the $\langle 0,0,0 \rangle$ point in world space (Figure 8), if you visualize yourself looking at the plane edge-on, it is simple to visualize this (Technologies, U).

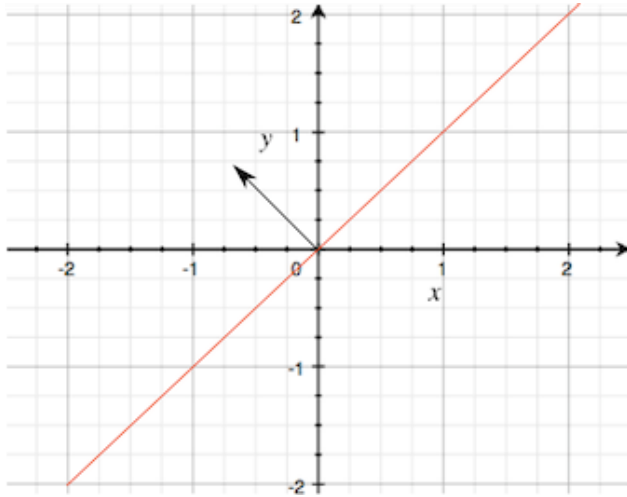


Figure 18 plane at points $\langle 0,0,0 \rangle$ in world space source:

Since the plane is a 3D GameObject with a mesh component, the UV coordinates of the mesh had to be modified. UV coordinates are used to map the textures onto the 3D objects. (Figure 18) is a code snippet from bounding.cs which accesses the mesh's vertices (mesh.vertices) and initializes an array 'uvs' to hold new UV coordinates for each vertex after retrieving the mesh component from the GameObject it is attached to. By taking the original UV x-coordinate (uvs_old[i].x) and halving the y-coordinate (uvs_old[i].y * 0.5f), it generates a new Vector2 for the UV coordinates. This effectively reduces the texture mapping for each vertex by 50% along the y-axis.

```

// Start is called before the first frame update
void Start()
{
    Mesh mesh = GetComponent<MeshFilter>().mesh;
    Vector3[] vertices = mesh.vertices;
    Vector2[] uvs = new Vector2[vertices.Length];
    List<Vector2> uvs_old = new List<Vector2>();

    mesh.GetUVs(0, uvs_old);

    for (int i = 0; i < uvs.Length; i++)
    {
        uvs[i] = new Vector2(uvs_old[i].x, uvs_old[i].y*0.5f);
    }

    mesh.uv = uvs;
}

```

Figure 19 code snippet from **Bounding.cs**

3.8.2 2D video visualization component

Implementing the 2D video became straightforward following the completion of the 3D video. For the 2D video, the project introduced two new planes to display the video in 2D, utilizing the methods from Nischita's 2019 tutorial. The project successfully played the video on these planes and applied the same texture used in the 3D video, as the video content remains consistent across both 2D and 3D implementations. The primary distinction was that the 2D screen rendered to a main camera, which targeted both eyes and was assigned to the default layer. The bounding.cs script (Figure 18) is used to map the textures onto the 3D objects by modifying the UV coordinates.

3.8.3 3D model visualization component

This student's supervisor took the 3D anatomical model from a patient's MRI scan, segmented it, and exported it as an OBJ file from MeshLab. We then imported this OBJ file into Unity and placed it in the scene. The kidney model, consisting of the kidney, tumour, veins, sinus, collecting system, and artery, was made up of several OBJ files, creating a comprehensive 3D anatomical kidney model. In Unity, an empty GameObject was created and named KidneyCollection and these OBJ models were attached as children. Each child object featured a mesh renderer and a box collider as components.

We applied textures of different materials to these objects for differentiation on 3D screens. For interaction, a TwoHandManipulation.cs script and a SceneContentAdjuster.cs were added as components to the parent GameObject, KidneyCollection. Within the TwoHandManipulation script's inspector settings on the parent GameObject, the manipulation mode was set to move, scale, and rotate, allowing users to manipulate the model as depicted in (Figure 19).

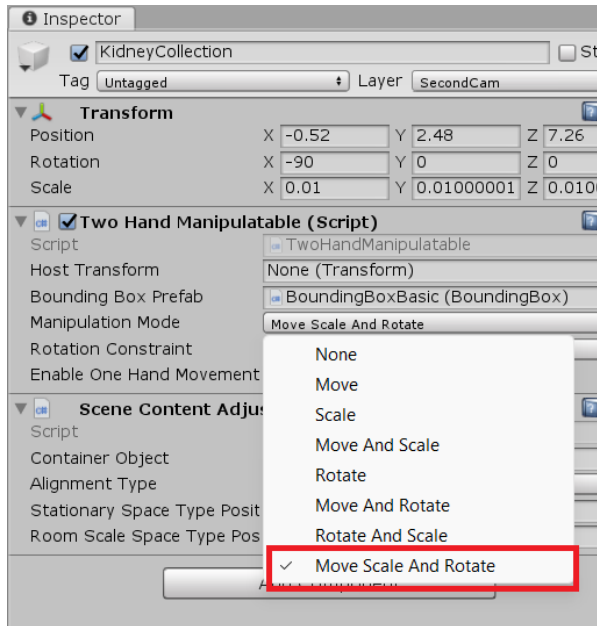


Figure 20 Manipulation mode

Within an MR environment, content is actively positioned and aligned by the Scene Content Adjuster script. By presenting objects at appropriate heights and locations based on the user's viewpoint, it improves the MR experience's realism and comfort. This script component has an alignment property type with the following settings: Use Preset Positions, Align with Head Height, and Align Preset X and Z With Head Height. As illustrated in Figure 20, the 'Align with Head Height' setting was manually selected for this project.

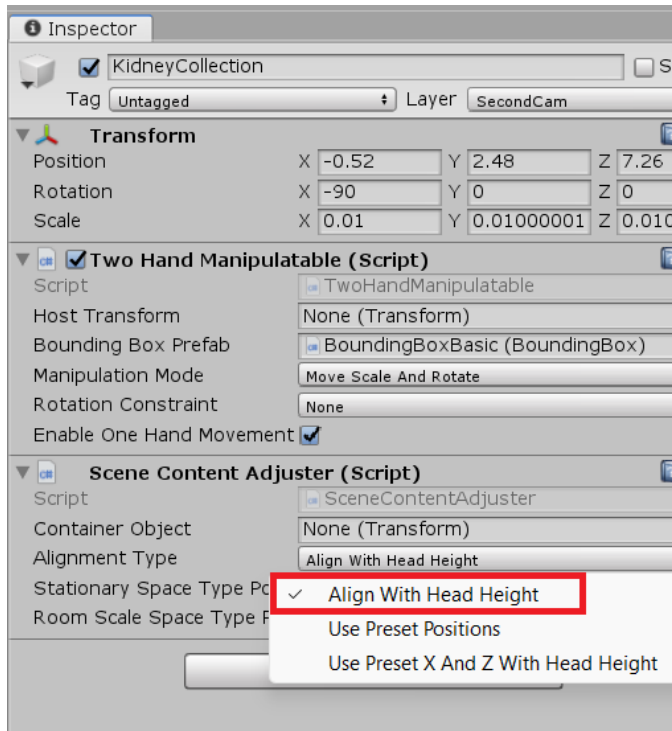


Figure 21 Alignment type

If the alignment type involves aligning with head height, it sets the y-coordinate of the content Position to the y-coordinate of the main camera (user's head height in this context), while preserving the x and z-coordinates of containerObject (Figure 21).

```

SceneContentAdjuster.cs
Assembly-CSharp
HoloToolkit.Unity.Boundary.SceneContentAdjuster
SetContentHeight()

84         yield return null;
85     }
86
87     if (alignmentType == AlignmentType.UsePresetPositions || alignmentType == AlignmentType.UsePresetXAndZWithHeadHeight)
88     {
89         if (XRDevice.GetTrackingSpaceType() == TrackingSpaceType.RoomScale)
90         {
91             containerObject.position = roomScaleSpaceTypePosition;
92         }
93         else if (XRDevice.GetTrackingSpaceType() == TrackingSpaceType.Stationary)
94         {
95             containerObject.position = stationarySpaceTypePosition;
96         }
97     }
98
99     if (alignmentType == AlignmentType.AlignWithHeadHeight || alignmentType == AlignmentType.UsePresetXAndZWithHeadHeight)
100    {
101        contentPosition.x = containerObject.position.x;
102        contentPosition.y = containerObject.position.y + CameraCache.Main.transform.position.y;
103        contentPosition.z = containerObject.position.z;
104    }
105    containerObject.position = contentPosition;
106
107    }

```

Figure 22 Scene content adjuster script

Finally, the ‘SecondCam’ is used to layer the model for HoloLens visualization. This integration makes sure the model is appropriately superimposed in the mixed reality environment of the HoloLens, enabling an engaging and interactive experience.

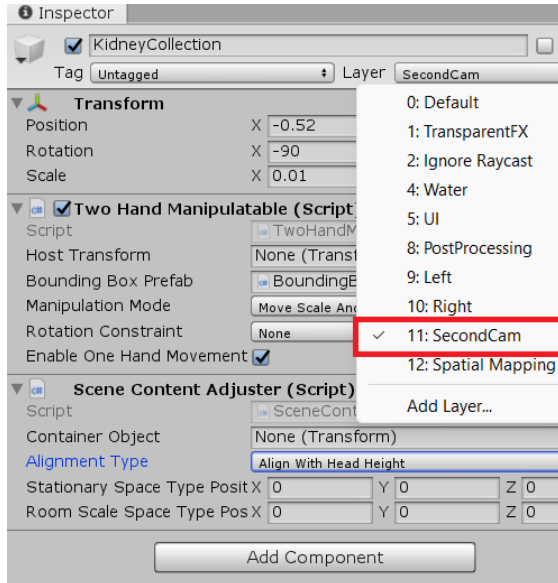


Figure 23 Layer for the parent GameObject

3.8.4 Holotoolkit

The following provide a detailed description on how this project actively utilized the components of the HoloToolkit.

3.8.4.1 Input Manager

A key component for managing input events in mixed reality apps developed for HoloLens and other MR/AR platforms is the Input Manager prefab found in the Holotoolkit for Unity. Its goal is to make handling different kinds of user inputs—such as voice commands, motion controller actions, and gestures—easier.

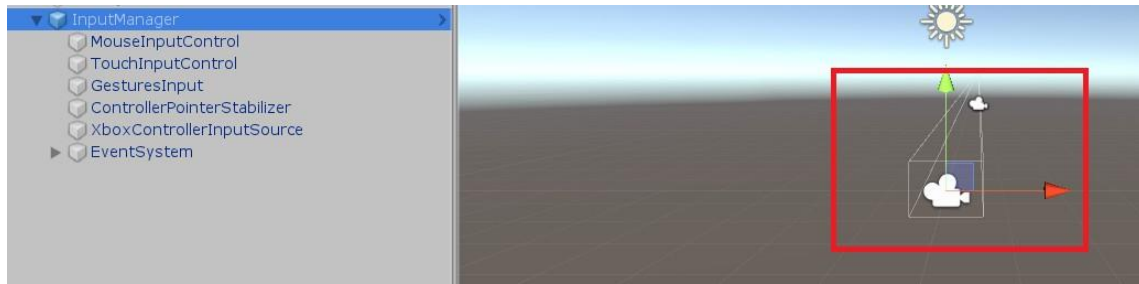


Figure 24 InputManager prefab in unity hierarchy and in the scene

The Input Manager is essential to the efficient operation of the grab-and-move interactions and two-hand 3D model manipulation. It serves as the primary system for understanding and processing user inputs necessary for these interactive features, including motion controller movements and gestures. By using the Input Manager, users can effortlessly manipulate the 3D model with both hands by having the application recognize particular gestures, such as pinching or grabbing. Its smooth conversion of these gestures into in-application actions guarantees a responsive and easy-to-use interface.

3.8.4.2 Cursor

A cursor provides continuous feedback based on where the headset believes a user's current focus is at a given time. The cursor instils user confidence during interaction. Usually, you place it at the point in the real world where the user's head gaze ray crosses a surface, be it a hologram or another object.

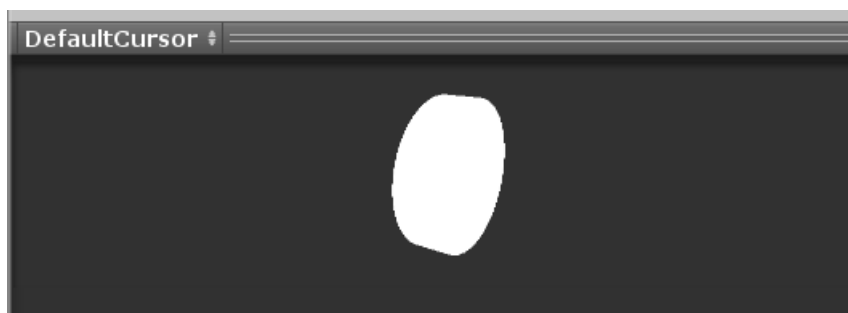


Figure 25 DefaultCursor prefab from Holotoolkit

Drag and drop the Default Cursor into the SimpleSinglePointerSelector component's Cursor field from the Inspector settings of the InputManager (Figure 21). By doing this,

the Default Cursor is designated as the main pointer interface and becomes capable of receiving input from the user.

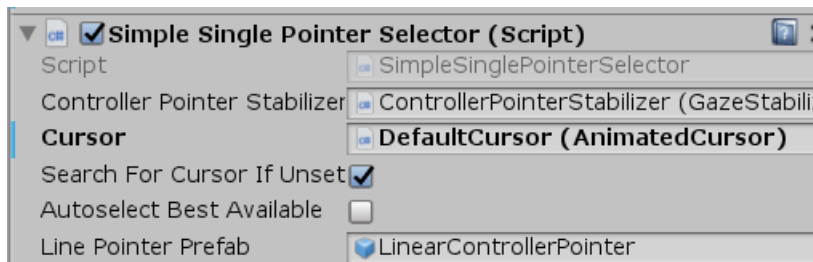


Figure 26 Default cursor assigned as pointer interface to Input manager.

3.8.4.3 Mixed reality camera

The Mixed Reality Camera Parent prefab is a key component in Unity's Holotoolkit this prefab's camera is meant to follow the head movements of the user. The Mixed Reality Camera Parent prefab features a Mixed Reality Camera actively configured with 'Depth Only' clear flags and culling masks targeting the 'SecondCamera' layer, enabling it to render objects assigned exclusively to this layer (Figure 22). This configuration is vital for rendering both 3D and 2D videos, as well as the 3D anatomical model effectively.

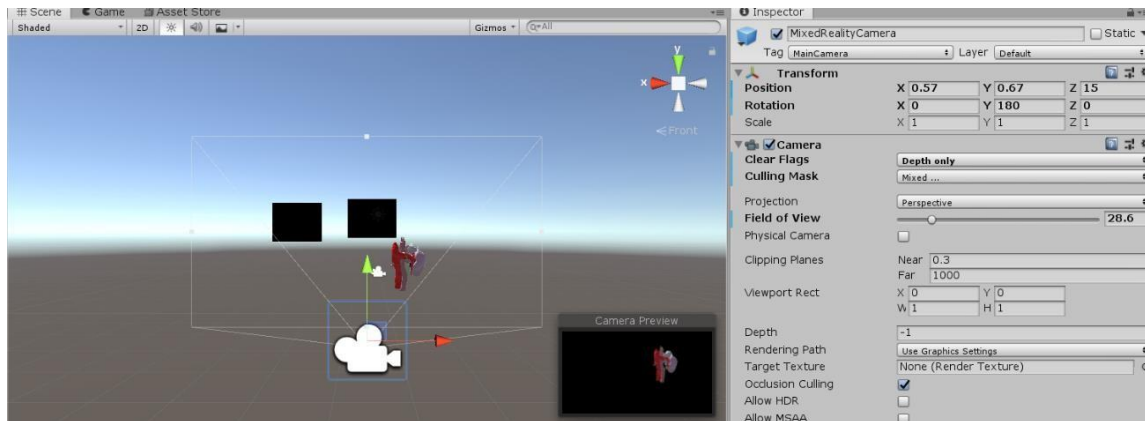


Figure 27 Mixed reality camera prefab

3.9 HoloLens gestures

Some of the basic interaction that are required in the project are indicated in the following sections.

3.9.1 Bloom gestures

One of the primary means of interaction for the first-generation Microsoft HoloLens was the "Bloom" gesture. It was a crucial component of the device's gesture-based navigation system, enabling users to return to the HoloLens Start menu or exit an application. The Bloom gesture is made by users by:

1. With your fist closed, begin.
2. Spread their fingers wide apart to open their hand.

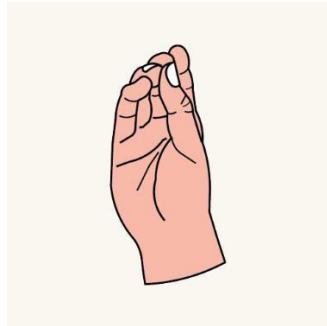


Figure 28 Bloom close (Source: Github)

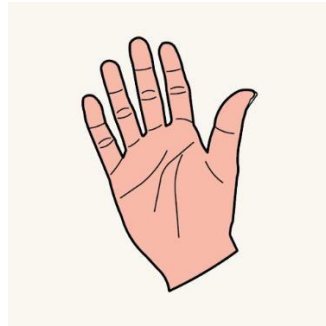


Figure 29 Bloom open (Source: Github)

3.9.2 Gaze and Air-tap gestures

In HoloLens, "gaze" refers to how the device tracks the user's gaze to enable virtual world interaction. It's an essential component of the HoloLens' and other mixed reality environments' user interface. To perform an air tap, start by raising your hand and positioning your index finger in the ready position. Next, pinch it with your thumb, and then lift it back up to complete the air tap motion. The most often used as input on the HoloLens (1st gen) is air tap.

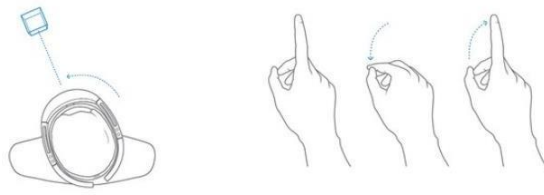


Figure 30 gaze and air-tap gestures (Source: Uzun, 2018)

3.10 User testing

The preparation and execution of the user evaluation study are covered in the following sections.

3.10.1 Pilot study design

A preliminary user study has been conducted to assess the recommended MR platform's usability. Seven students in all, with different ages, educational backgrounds, and levels of experience with AR/VR technology, were asked to participate in this research. Every student received an overview of the platform along with a detailed explanation of its features. Following that, each user had ten minutes to engage with and investigate the features of the visualisation. The students responded to the questionnaire (Figure 23, 24) at the conclusion of the study. The purpose of the survey is to seek input from the participants and evaluate the usability of the suggested visualisation platform. It is divided into three sections namely: Section 1 - Alignment and interaction, section 2 - Educational value, section 3 - general feedback.

Participant questionnaire

Section 1: Alignment and Interactions

Q1: how would you rate the ease of aligning the kidney model with the tumour in the application?

☐ 1 (Not easy)

☐ 2

☐ 3

☐ 4

☐ 5 (very easy)

Q2: How clear was the visualization of the kidney and the tumour in both the 3D and 2D video screens?

☐ 1 (Not clear)

☐ 2

☐ 3

☐ 4

☐ 5 (very clear)

Q3: Were you able to clearly see the blood vessels in the video?

☐ 1 (not clear)

☐ 2

☐ 3

☐ 4

☐ (not clear)

Q4: Please rate the effectiveness of the 3D model in enhancing your understanding of the procedure:

☐ 1 (not effective)

☐ 2

☐ 3

Figure 31 Page 1 of user questionnaire

☐ 4

☐ 5 (very effective)

Q5: Please rate the effectiveness of using two hands to interact with the 3D model compared to using a single hand for the video screens interaction.

☐ 1 (less effective)

☐ 2

☐ 3

☐ 4

☐ 5 (very effective)

Q6: Which did you find more helpful for understanding the anatomy and the surgical procedure - the 3D video screen or the 2D video screen? Please explain your choice.

Reason:

Q7: Describe your experience interacting with the 3D anatomical patient model. What worked well, and what could be improved?

Description:

Section 2: Educational Value

Q8: How effective do you think this application is for educating users about surgical cases.

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☐ 5 (effective)

Q9: Compared to traditional learning methods (like textbooks or lectures), how effective do you find this mixed reality application in providing educational content?

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

Figure 32 Page 2 of user questionnaire

☐ 5 (effective)

Q10: What improvements or additional features would you suggest to enhance the educational value of this application?

Suggestions:

Section 3: General Feedback

Q11: What was your overall experience using this application? Feel free to include any additional comments or suggestions.

Feedback:

Q12: In which of the following areas would you recommend using this platform?

☐ Practice Training

☐ Medical and Patient Education

☐ Surgery

☐ Rehabilitation

☐ Patient Treatment

Figure 33 Page 3 of questionnaire

Respondents can indicate their degree of agreement or satisfaction on a five-point Likert scale by answering questions 1, 2, 3, 4, 5, 8 and 9. Conversely, Questions 6, 7, 10, 11 and 12 show a flexible form that permits participants to freely share their experiences and ideas in great detail. Based on this metric, a score of 1 denotes the least likelihood of support, whereas a score of 10 indicates the greatest likelihood of strong support for the application.

It was requested of the users to rate the application's ease of use in aligning a kidney model with a tumour, the clarity of the visualization, the visibility of blood vessels, whether they preferred 3D or 2D video screens, and how they interacted with the 3D anatomical patient model. In Section 2, questions about the platform's usefulness as an educational use were covered, including how well it taught surgical cases, how it compared to other traditional learning approaches, and what changes could be made to increase its educational value. Section 3 covered the users' overall experience with the platform and their likelihood of recommending the platform for clinical cases.

3.10.2 Analysis of questionnaire answers

The responses we received are analysed in this section. Figure 26 shows the average and individual scores for Questions 1, 2, 3, 4, 5, 8 and 9 where each student marked their

responses on a scale from 1 to 5. Additionally, it showcases the opinions students provided for Questions 6, 7, 10, 11 and 12.

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Participant	Ease of Aligning Model (1-5)	Clarity of Visualization (1-5)	Blood Vessels (Clear/Not clear)	Effectiveness of Two-Hand Interaction	Preference (3D/2D Screen)	Understanding in procedure (1-5)	Interaction with 3D Model	Effectiveness in Education (1-5)	Compared to Traditional Methods (1-5)	Suggestions for Enhancements	Overall Experience	Area of recommendation
1	A	4	5	5	3	3D Screen	4	Good, needs smoother movement	5	5	Clearer instructions	Positive	Practice Training
2	B	3	4	5	4	2D Screen	5	Challenging, technical glitches	3	4	Better software stability	Mixed	Practice Training
3	C	4	5	4	5	3D Screen	5	Easy interaction, improvement needed	5	5	Real-time axis information	Very Positive	Medical and Patient Education
4	D	2	3	5	4	2D Screen	5	Difficult, alignment issues	2	3	More intuitive controls	Average	Patient Treatment
5	E	5	5	5	5	3D Screen	5	Very intuitive, smooth operation	4	4	More anatomical details	Excellent	Medical and Patient Education
6	F	3	4	5	3	3D Screen	5	Moderately easy, some lag	3	2	faster response	Good	Medical and Patient Education
7	G	3	4	4	3	3D Screen	5	Great experience	5	5	Voice commands, detailed textures for model	Good	Medical and Patient Education
8	Average	3.42	4.28	4.71	4	N/A	4.85	N/A	3.85	4	N/A	N/A	N/A

Figure 34 Questionnaire scores given by the students.

Participants rated the ease of aligning the 3D model between 2 and 5, with an average of 3.42, showing varied levels of user comfort. They generally viewed the clarity of visualization positively, as reflected by the high average score of 4.28. While opinions on the visibility of blood vessels were mixed, the effectiveness of two-hand interaction received favourable feedback, averaging at 4. Preferences split between 3D and 2D screens, highlighting diverse user comfort levels. All of the participants agreed that the 3D model will help in understanding of the procedure and their experiences with the 3D model ranged from smooth and intuitive to technically challenging.

The application's effectiveness in education scored an average of 3.85, demonstrating its competence over traditional methods, which similarly averaged a positive score of 4. Participants were inclined to recommend the application, with an average likelihood of 8.14 out of 10. Overall, the application garnered a positive reception, coupled with constructive suggestions for clearer instructions, enhanced software stability, and more intuitive controls to refine the application further.

We can see whether participants thought the 3D or 2D presentation was more effective for educational use by looking at the "Preference (3D/2D Screen)" and "Effectiveness in Education (1-5)" columns. Given below is a bar graph (Figure 27) to analyse the result of

this discussion. When compared to the 2D screen, the 3D screen was generally thought to be more effective for understanding the procedure, with an average effectiveness rating of 4.4.

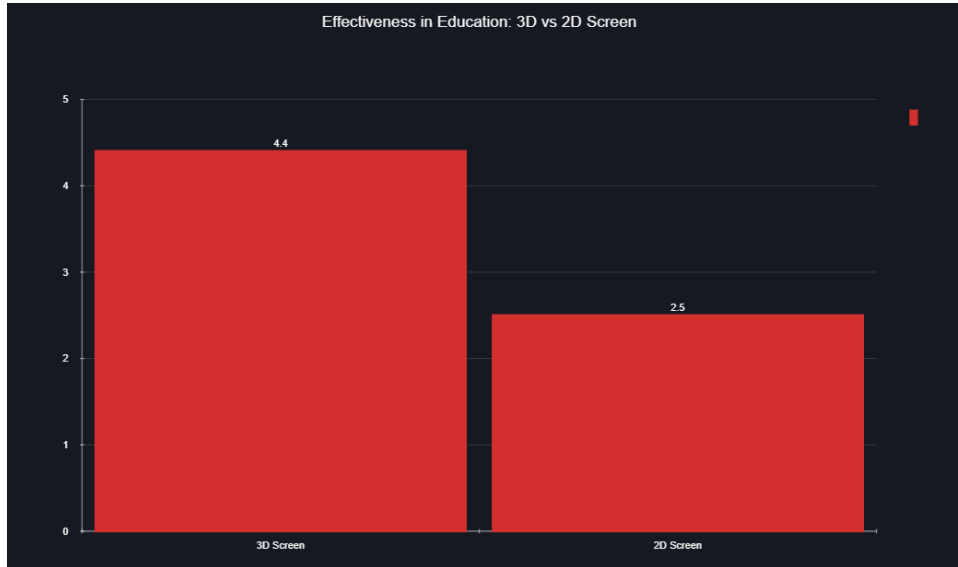


Figure 35 Bar chart for understanding procedures using 3D and 2D screens.

Question 12 looks into the efficacy of the suggested platform for various medical specialties. "In which of the following areas would you recommend using this platform". As shown in (Figure 36) 4 out of the 7 participants found the application to be useful in Medical and patient education. The second most suitable area is Practice training. This implies that the tool is especially helpful in educational settings, possibly for instructing medical students or providing patients with explanations of procedures.

We are not attempting to address any particular clinical issue in this paper. Rather, this is an initial phased demonstration of our visualization platform's suitability for training and instructional purposes and perhaps aid in surgical procedures in a medical setting.

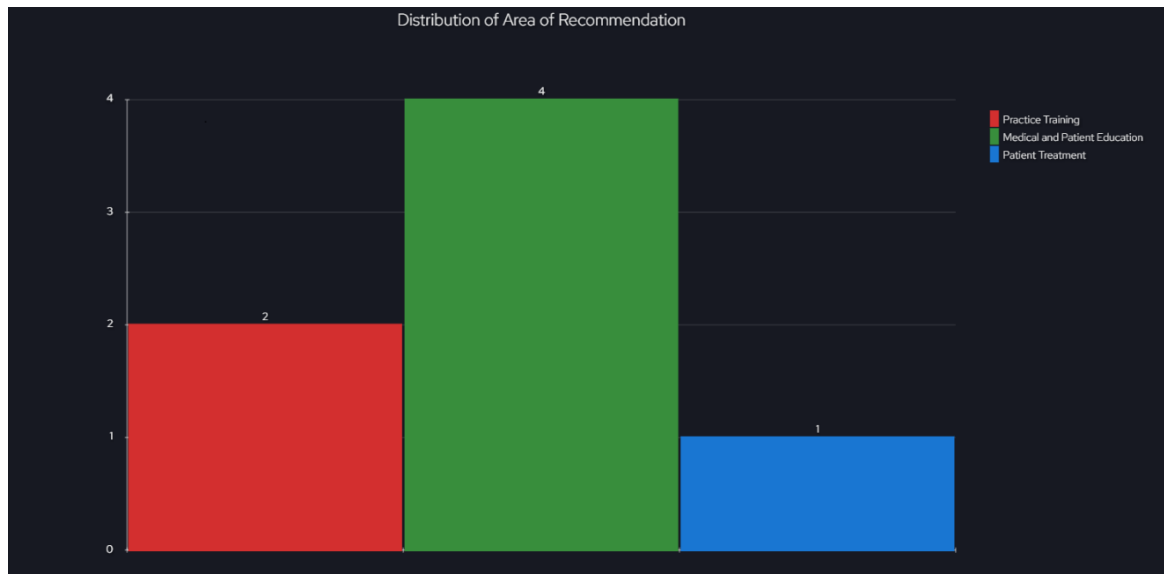


Figure 36 bar graph for area of recommendation

4 Results

This chapter outlines the project's results, particularly highlighting the mixed reality application in action. It shows images and provides explanations of some of the key outcomes, along with relevant diagrams.

4.1 Platform Images

The following sections provide images from the HoloLens device, Unity, and user testing.

4.1.1 Unity Images

The following sections present images from Unity to display the results of the asset presentation and the scene within Unity.

4.1.1.1 Unity Scene

A Unity scene with a mixed reality camera preview is shown in (Figure 36). The scene includes a 3D model and two screens, one on the left and one on the right, for the 2D and 3D respectively. The mixed reality camera renders both the 2D screen and the 3D model in the camera preview. The 3D screen and other visual elements, which are not visible in the camera preview, are displayed by the HoloLens device.

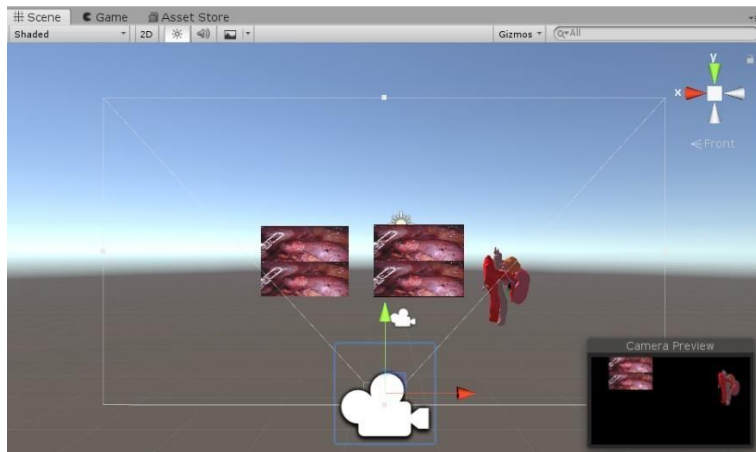


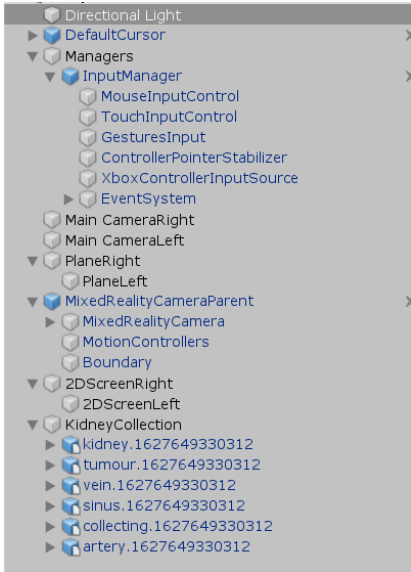
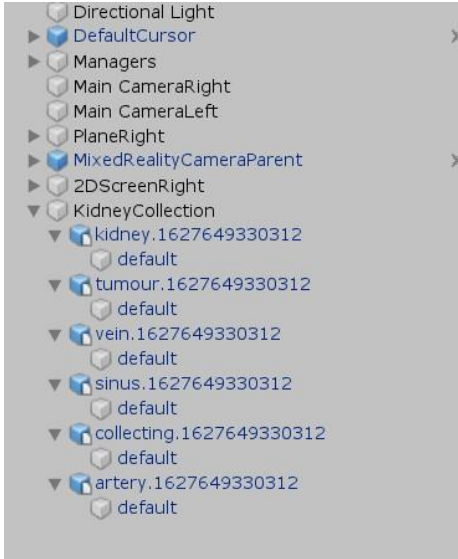
Figure 37 Unity Scene display with camera preview

4.1.1.2 Unity's Hierarchy

The hierarchy of projects in Unity is shown in the table below, which also shows where GameObjects are stored. It consists of the following five parent GameObjects:

KidneyCollection, 2DScreenRight, PlaneRight, MixedRealityCameraParent, and Managers. Prefabs from the Holotoolkit package are the DefaultCursor, InputManager, and MixedRealityCameraParent. The .obj files, representing the kidney, tumour, vein, sinus, collecting system, and artery, come with a default model, allowing for the addition of components. Managers and KidneyCollection function as two empty GameObjects. Meanwhile, Main CameraLeft, Main CameraRight, PlaneRight, PlaneLeft, 2DScreenRight, and 2DScreenLeft are the project's primary GameObjects.

Table 2 Unity's Hierarchy

 Figure 38 Project Hierarchy	 Figure 39 Hierarchy extended for KidneyCollection.
--	--

4.1.1.3 Unity’s project window

The Project Window can be repositioned to suit your preferred workflow, although it normally appears at the bottom of the Unity Editor.

The folder hierarchy on the left and the asset view on the right make up its two primary sections. The asset view shows the contents of the selected folder, while the folder

hierarchy shows the arrangement of the folders in your project. There are 5 folders which are created namely, KidneyObjects, Materials, Scenes, Scripts, VideoPlayer (figure 38). A detailed categorization of this project’s assets is shown in table 2.

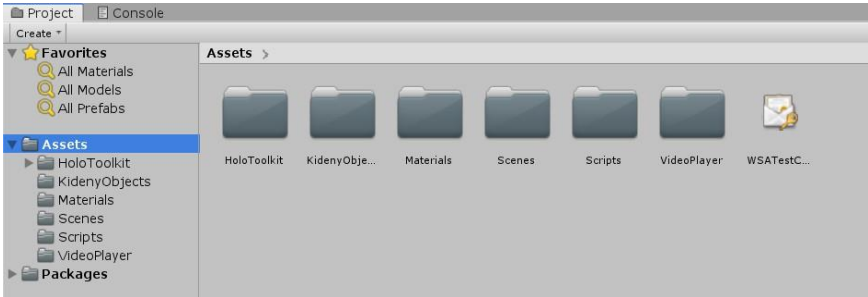


Figure 40 project's assets organisation

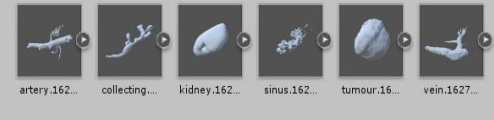
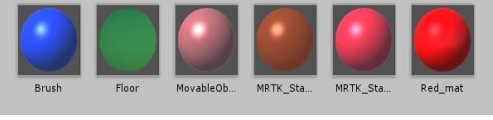
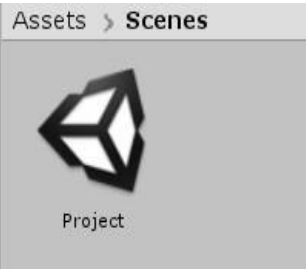
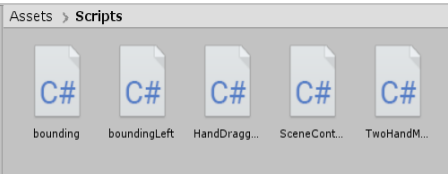
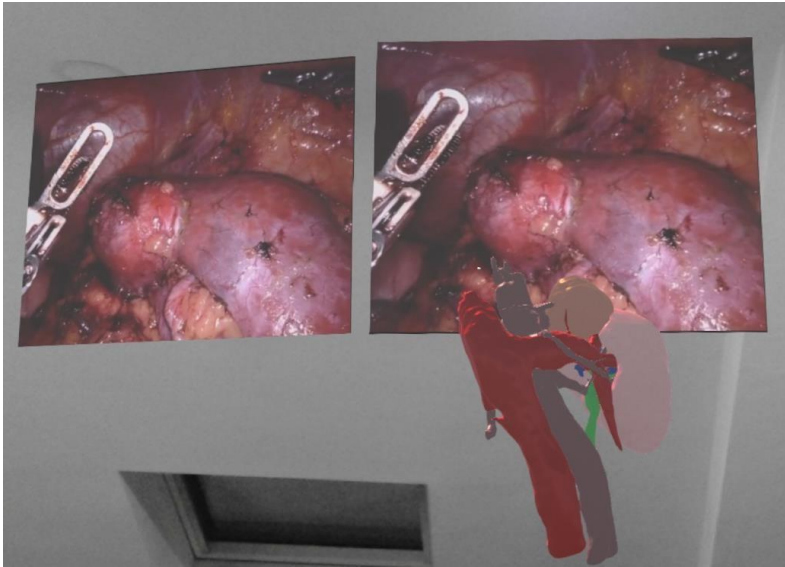
Assets > KidneyObjects 	Assets > Materials 
Assets > Scenes 	Assets > Scripts 
Assets > VideoPlayer 	

Table 3 Assets categorisation

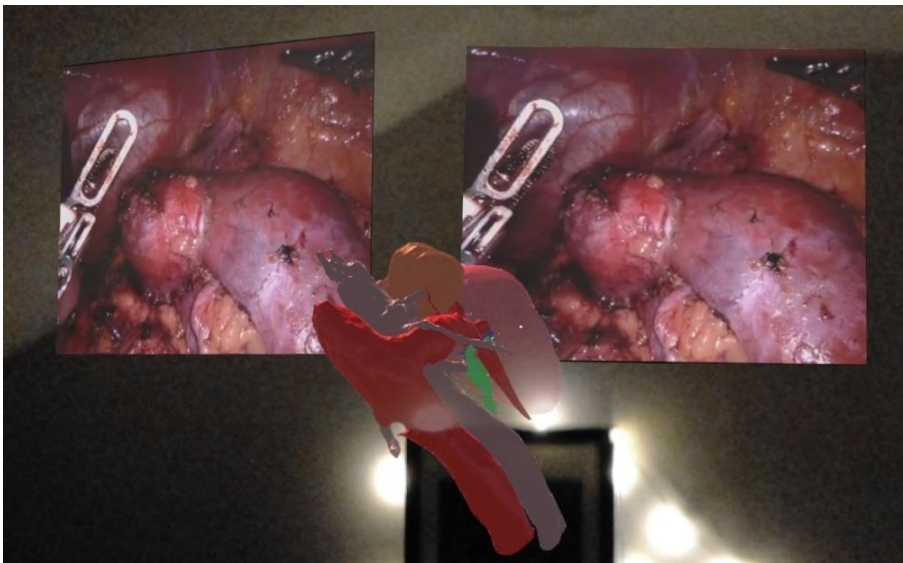
4.1.2 HoloLens Images

These series of photos show the user lining up the 3D anatomical model with their favourite 2D or 3D screen, along with where the cursor is located on the screen and the

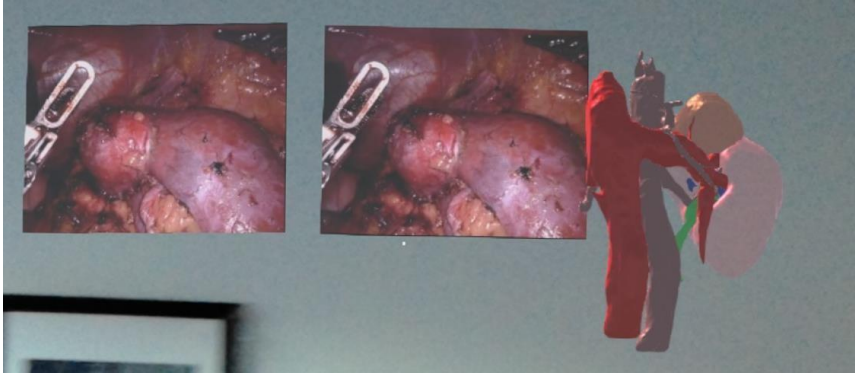
model. It also has pictures that are limited to the model and screens. These images shows the positioning of the model in relation to the screens by participants A through G.



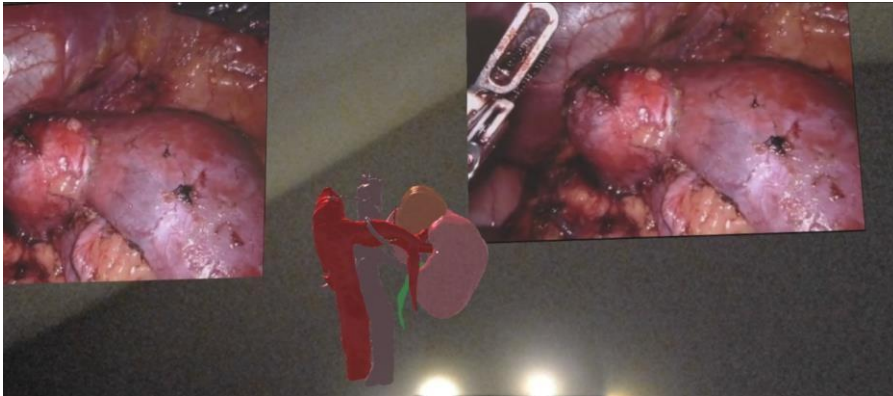
Participant A



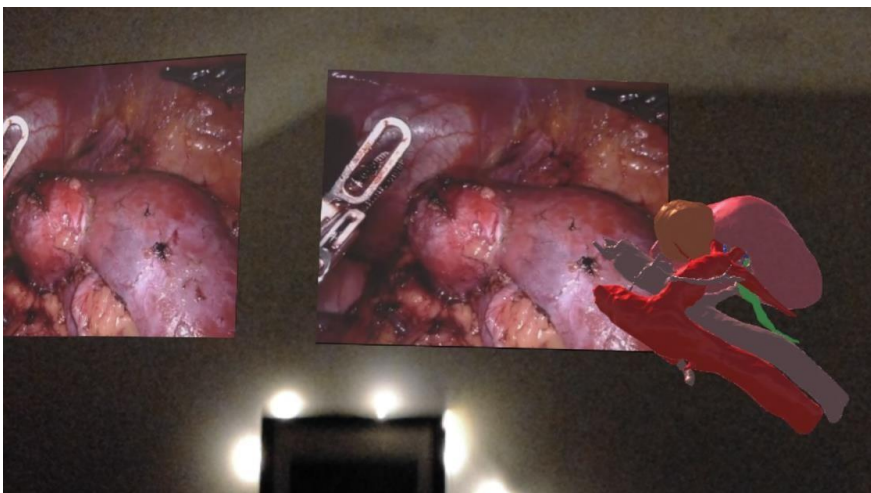
Participant B



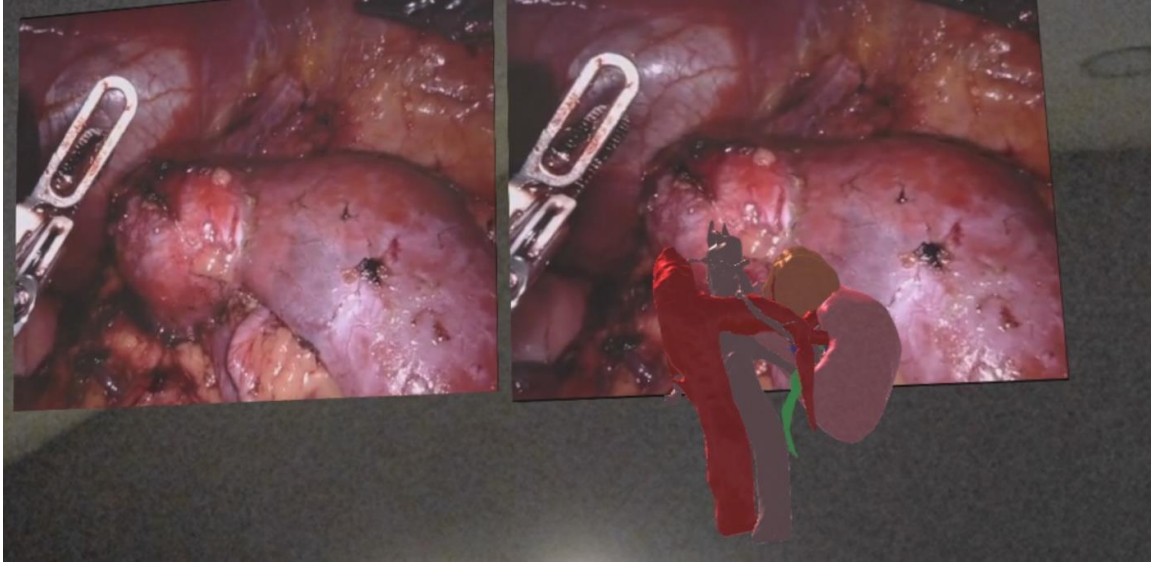
Participant C



Participant D



Participant E



Participant F



Participant G

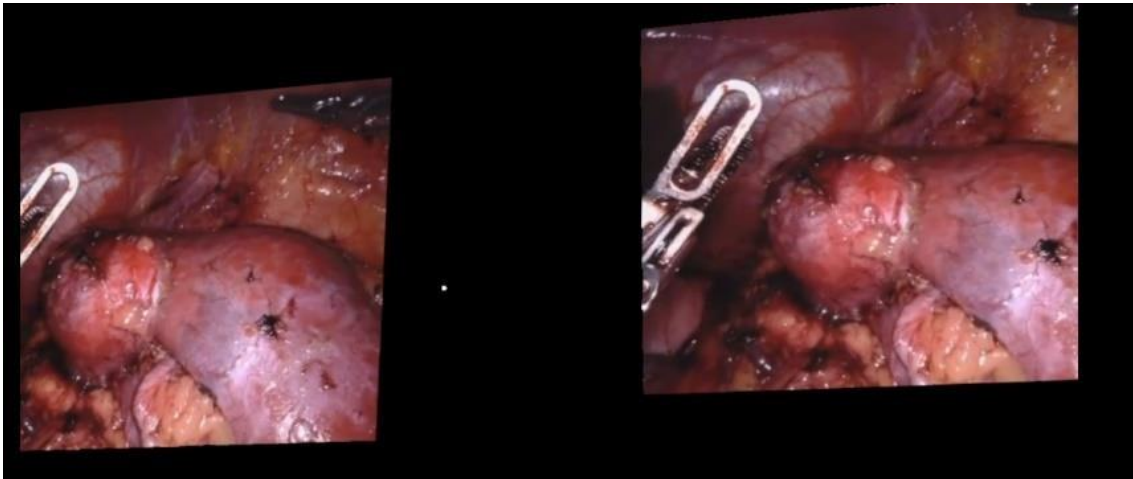


Figure 41 2D screen on the left and 3D screen on the right.



Figure 42 Anatomical model with textures

4.1.2.1 Cursor Interactions with the visualisation components

The images below demonstrate how the cursor interacts with the 2D screen. In (Figure 42), the cursor targets the screen for interaction, and the component recognizes this action. In (Figure 43), the user pinches the cursor by raising the index finger and using the thumb to pinch, resulting in the movement of the screen

Table 4 cursor interaction with 2D screen

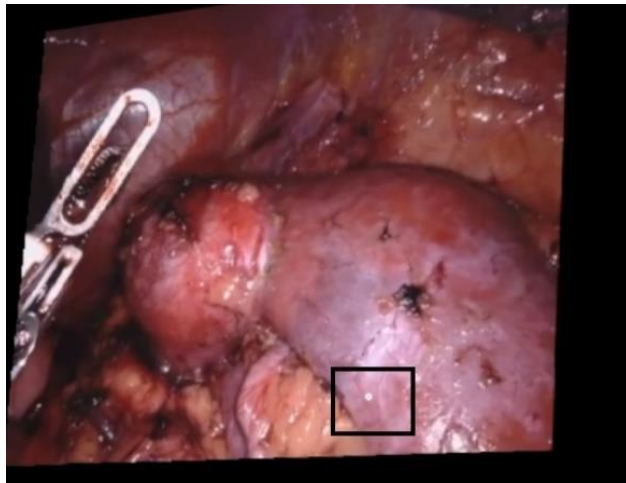


Figure 43 Cursor pointed on the screen.

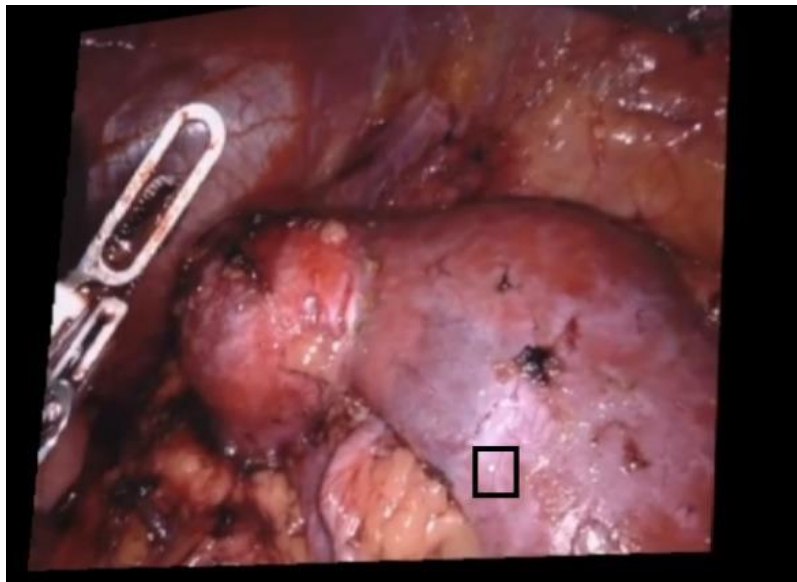


Figure 44 Cursor pinched

The table below shows the same actions of the cursor for the 3D screen.

Table 5 Cursor interaction with 3D screen



Figure 45 cursor pointed on the 3D screen

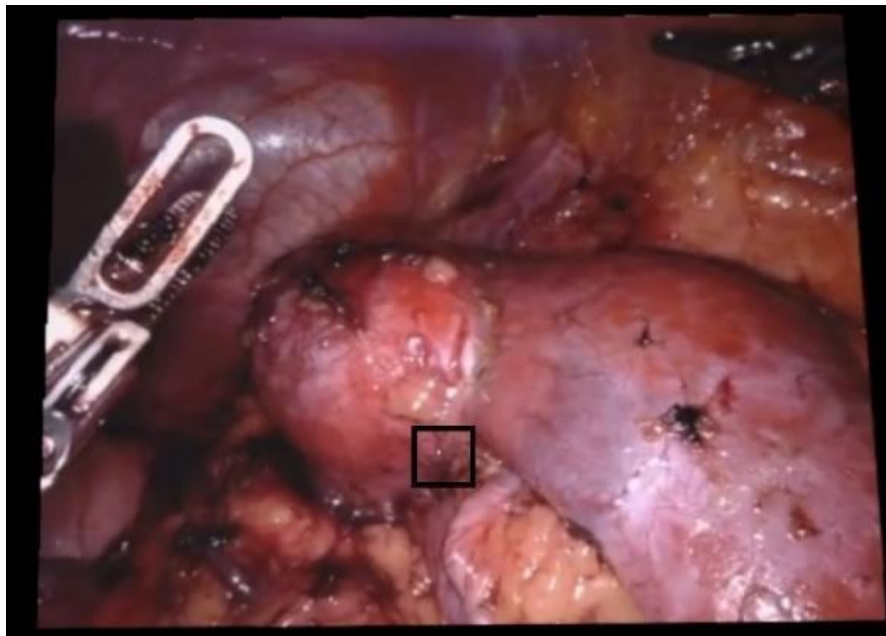


Figure 46 cursor pinched for interaction

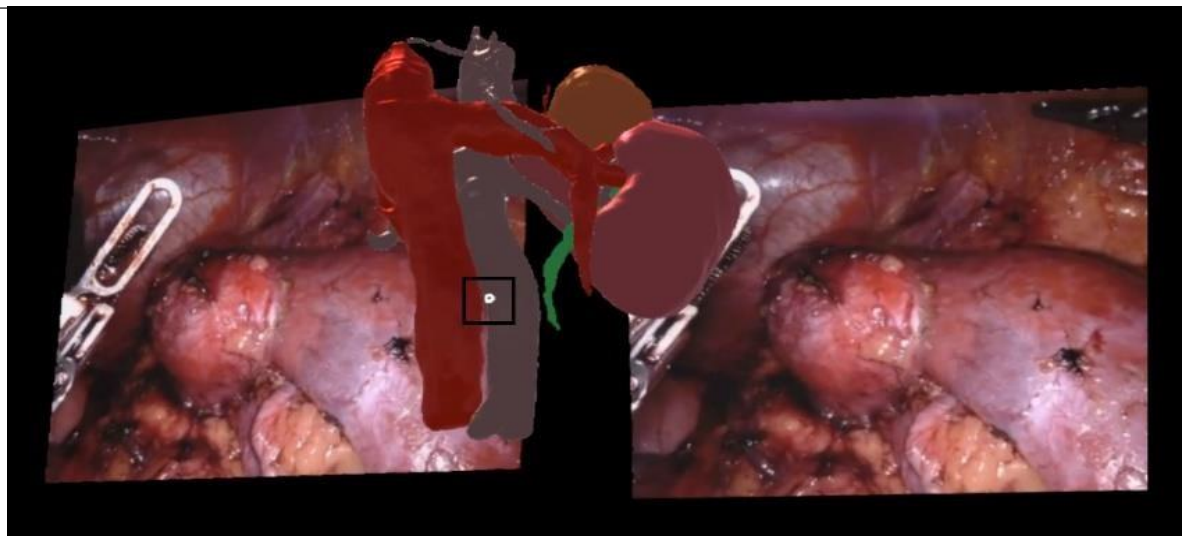


Figure 47 Cursor pointed on the model.

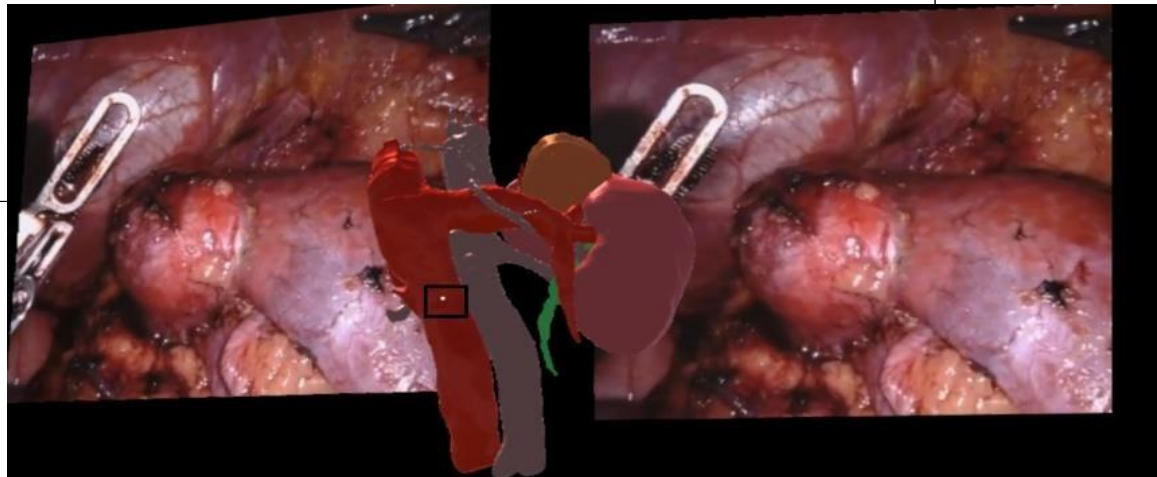


Figure 48 cursor pinched for interaction

Table 6 Cursor interaction with 3D model

4.1.3 User testing Images

The following images shows the mixed reality application tested with the HoloLens 1 at this author's home. They illustrate the project's objectives, including two-hand interactions and the visibility of the 3D screen, being achieved.





Figure 49 More pictures of the user testing

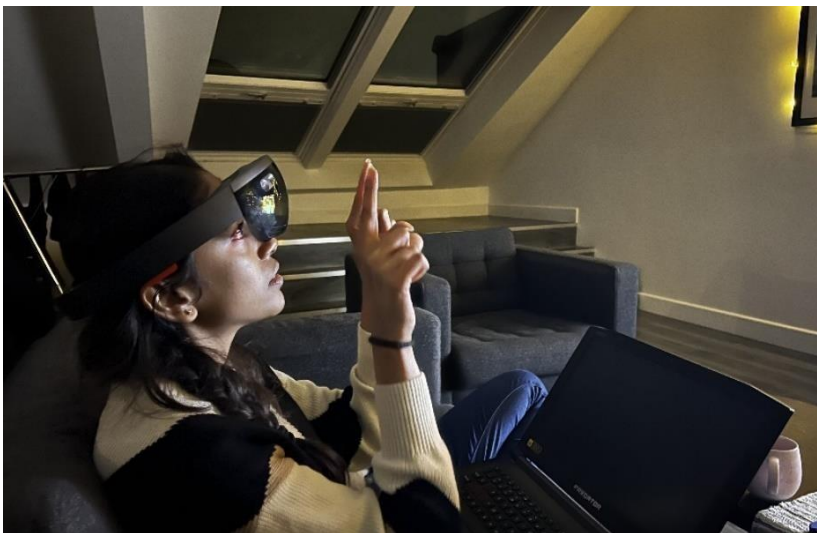


Figure 50 One participant tested the HoloLens while seated.

4.2 Diagrams

The following flowchart diagram and UML sequence diagram were produced from Draw.io. (Draw.io)

4.2.1 Gaze gesture system

A flowchart explaining the "Gaze gestures in HoloLens" is shown (Figure 41). "User's Line of Sight" is where the flowchart starts, highlighting the fact that looking at the HoloLens is the main way to interact with it. Following that, the flowchart divides into two parallel sections: "Head Tracking" and "Eye gaze".

These elements serve as the foundation for the HoloLens' interpretation of head and eye movements, which enables more accurate and natural interactions between the user and the virtual environment. This allows the user to select and manipulate objects or initiate specific commands.

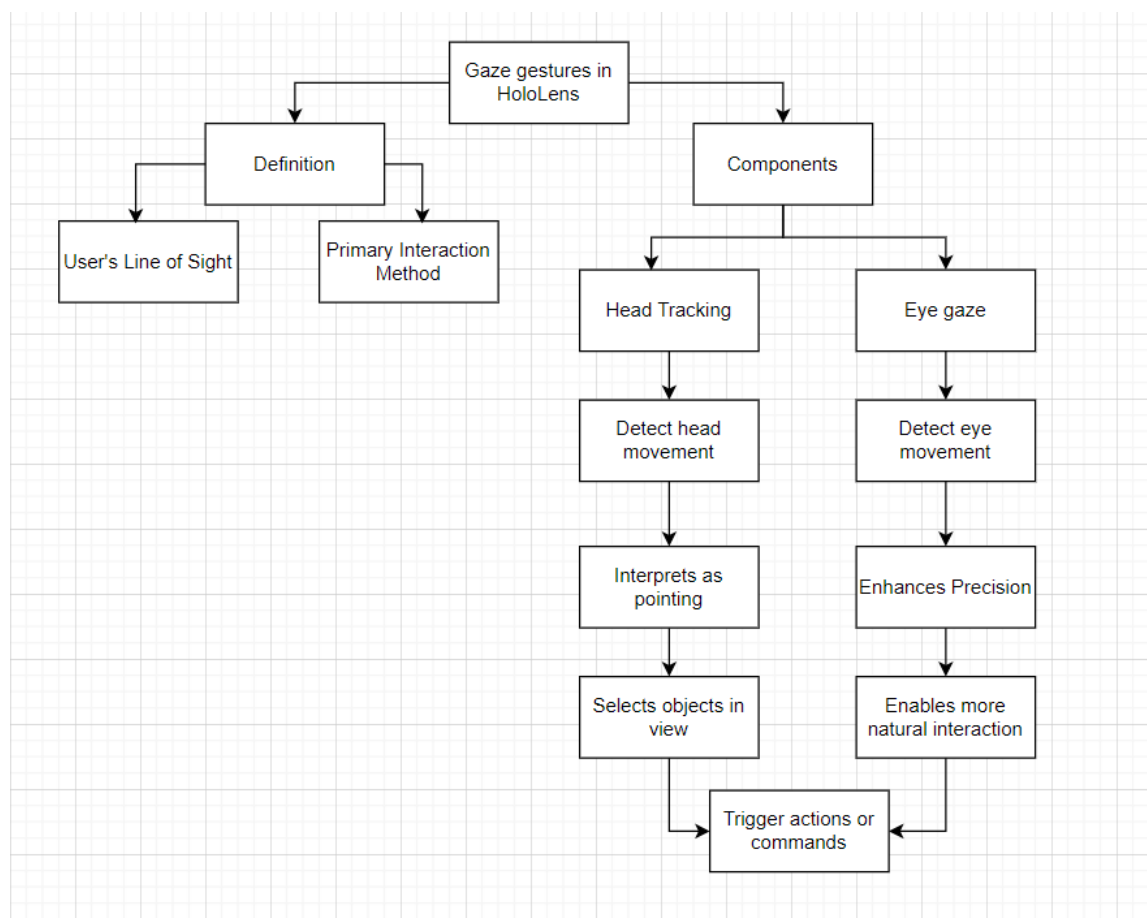


Figure 51 Gaze gesture system flowchart (produced from: Draw.io)

4.2.2 UML Sequence diagram for grab and move interactions.

According to the sequence diagram (Figure 42), the user signals the HoloLens system to begin an interaction by making a grab gesture to pick up a virtual object. Once this action is detected, the system confirms the grab and notifies the user that the action was accomplished. After that, the user moves their hand, causing the system to adjust the interactive visual component's position. The user continues to manipulate the object in this interaction loop, and the sequence is completed when the user releases the gesture, and the system stops updating the object's position.

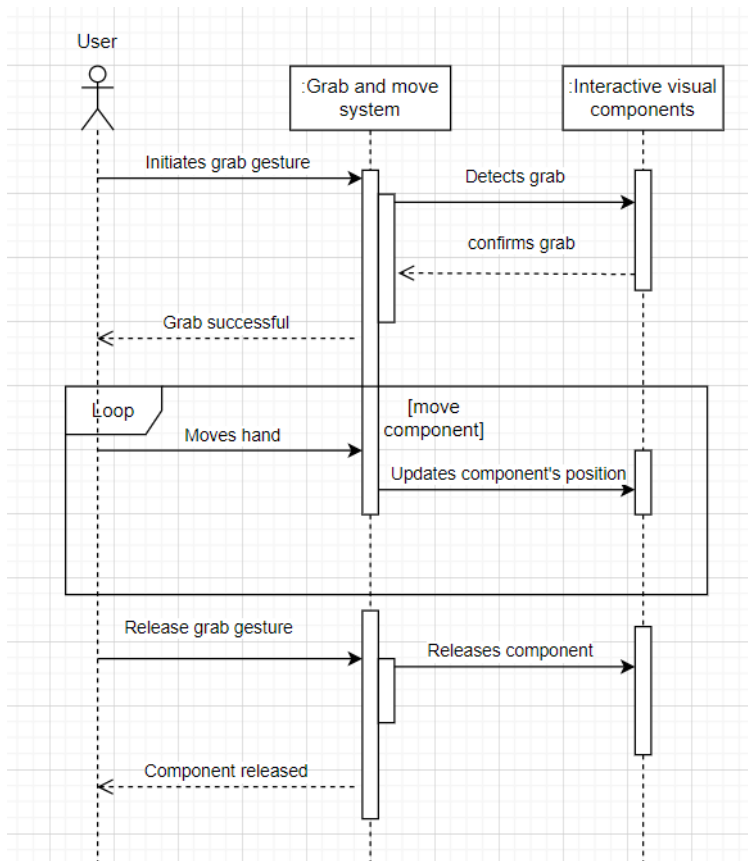


Figure 52 Sequence diagram for grab and move interaction. (produced from: Draw.io)

4.3 Summary of key results

This project demonstrated in response to the research questions that creating a 3D stereoscopic display in HoloLens 1 is not only possible, but it can also be enjoyable. Furthermore, it showed that both the interactions and the device are intuitive and user-friendly.

This project also revealed that the HoloLens 1, while not cheap, provides an incredibly immersive mixed reality experience at a price appealing to both developers and gaming enthusiasts. The hardware responds well to physical movements, offering a seamless view into a 3D environment. According to our evaluation, almost all users reported that it was responsive, immersive, and generally comfortable. Unity's robust platform also simplifies the development process, making it more accessible for creating interactive mixed reality applications that are both practical and enjoyable.

5 Discussion

This chapter compares the application produced with the original objectives and details how well it achieved them. It provides specifics on how the final project deviated from the plan and shares some key lessons learned and recommendations.

5.1 Overall objectives

Reviewing the key results from this project, we will address each research question one by one, as outlined in Section 1.3:

Research question1: *Does 3D visualisation of the video help in understanding the procedure?*

The project not only succeeded in the technical implementation of a 3D video but also set the stage for subsequent clinical evaluation. Testing by clinicians to determine the video's effectiveness in enhancing procedural understanding is an anticipated next step, offering a promising opportunity for further validating the project's educational impact.

Research question2: *Does 3D aligning of the patient anatomy with a surgical scene help in understanding the procedure better?*

An average clarity score of 4.28 out of 5 was obtained from the user testing analysis, suggesting that participants found the visualization to be quite clear. Additionally, they found the 3D model to be very helpful in understanding the procedure, giving it an average helpfulness score of 4.85. Participant evaluations and comments testify to the fact that procedural understanding is improved by 3D rendering of patient anatomy. The next crucial phase involves clinician testing, which will provide critical insights into the clinical efficacy and applicability of the 3D rendered models in actual medical settings.

Research question3: *Does giving the ability to interact with medical data be helpful in medical setting?*

Although there is room for improvement, the data suggests that the participants valued the features of the platform, with an average ease of alignment score of 3.42 and a clarity score of 4.28. The model's average score of 4.85, which is exceptionally high, suggests that it has a strong positive influence on the understanding of spatial relationships. The

importance of these capabilities in facilitating understanding is further supported by the diverse comments made by participants about their interactions with the 3D model. These comments ranged from the need for smoother movement to experiencing intuitive operations.

The data shows that participants generally found the ability to interact with the model and the clarity of the visualizations helpful for conveying the 3D relationships of anatomical structures.

Research question4: *How can we involve this visualization platform in the healthcare setting to provide improved learning over traditional approaches?*

Despite varying effectiveness scores, the visualization platform includes two-hand interaction features that hold promise for a more engaging learning experience than conventional methods. High ratings for both education effectiveness (average 3.85) and model helpfulness (average 4.85) imply that the platform provides an enhanced learning experience over more conventional approach.

The bar graph in (figure 36) clearly recommends the use of this platform in one of the medical areas.

Research question5: *How can we use visualization components such as 3D anatomical model and 2D/3D screens to benefit clinicians and novices?*

The platform has integrated both 2D and 3D visualization components, demonstrating how effortlessly aligning the model with the 3D screen clarifies the blood vessels' scene. With the anatomical model's assistance, clinicians can easily visualize and comprehend the intricate structures. Clearly, these visualization components provide significant benefits to clinicians, enhancing their ability to understand and interpret complex medical information.

All things considered, this project has shown that it is feasible to use Hololens1 to create a fun responsive, and immersive mixed reality application. The results of the user evaluation show that providing users complete control over the visualization components at their fingertips helps them train more successfully, make better decisions, and plan

procedures more clearly. All of these benefits add up to a less confusing and more enjoyable user experience.

In the future development stages of this platform, we will conduct extensive testing with clinicians to refine and enhance this application's functionalities.

The subsequent sections of this chapter will delve more deeply into the difficulties that developed during the project, offer insightful observations, and examine the various choices that were made during this work.

5.2 Known issues with HoloLens.

If the USB connection to the system is not secure, the HoloLens application deployment via Visual Studio via USB will fail. Additionally, after connecting the HoloLens device via USB, make sure to turn it on. This problem can be very discouraging when it comes to deployment.

Sometimes, when deploying from Visual Studio, files like `Assembly-CSharp.dll` or `.csproj` files are not found. In the event that this occurs, remove the prior build folder, make a new one, build using Unity, and then deploy using Visual Studio.

5.3 Recommendations

As you develop this platform, I recommend that future developers familiarize themselves with the different versions of Unity to find the one that best suits their implementation requirements. Set up the platform early because you might discover that the platform lacks the necessary packages for your implementation. This initial setup delayed my progress by a few days. Once your environment is ready, ensure you match the mixed reality toolkit package with your game engine version. It's crucial to strike a balance between your interactive elements and the game object's components.

Be aware that large `.obj` files and high-resolution videos or images can cause lagging, which can frustrate users during testing. I learned to always back up the Unity scenes or make a backup of all the project files in the local machine. Understanding the offerings of the mixed reality toolkit packages early can greatly enhance your creativity.

Avoid a mistake I made, create a separate build folder instead of building your project in the project's root directory. The level of interaction available in HoloLens 2 appears to be fascinating and ideal for a mixed reality application, though I have yet to experience the HoloLens 2 device myself.

I discovered that deploying to HoloLens over USB was the simplest method. You just need to connect your HoloLens device to your system and configure the build settings in Unity and Visual Studio accordingly.

5.4 Summary of the lessons learnt

Working on this project gave me the chance to spend a lot of time with Unity and broaden my knowledge of it, which was a worthwhile experience. Using the HoloLens 1 was an amazing experience for me because I had never used any AR/VR technology before. I gained a lot of knowledge about its revolutionary effects and current applications in many different fields.

The development process went smoothly without any major issues, except for the challenge of time management. The project was kept on schedule by using a Gantt chart. The Gantt chart helped us stay on track to meet our milestones because it's very easy to underestimate the amount of time required.

One area that could have been smoother was the rendering of the visual components. This project required multiple cameras to display 3D and 2D screens, along with a 3D anatomical model. Coordinating a camera to render all these components was complex, but, fortunately, everything came together, and the implementation was successful.

The literature review was worthwhile and fascinating, but at times it was difficult to combine the data into a compelling story.

This was my first time conducting user testing successfully. I aimed to make the questions clear, especially since the respondents were students. If I were to do this project again, I would reach out to clinicians earlier to schedule their evaluation of the project.

6 Evaluation, Reflections and Conclusions

This chapter evaluates the entire project, draws conclusions, and summarizes the experience gained. It also offers suggestions for further work and reflects on the overall experience.

6.1 Project management

I am pleased with the outcome of this project. Allocating generous time for tasks allowed me to conduct research on different approaches to problems and to solve unexpected issues without derailing the overall timeline. Choosing the appropriate Unity version and the compatible Holotoolkit version for the available HoloLens 1 device was the most critical decision. Without this choice, the project would not have reached completion successfully.

6.2 Future work

Doing clinician and novice assessments of this platform will be a key future task. Improved interactive controls, such as a slider for playing videos and more interactions like voice commands to replace hand gestures, could be beneficial for the platform. Optimizing latency reduction will improve user experience. The platform's application in clinical scenarios may be expanded by enhancing stability through the resolution of technical issues and adding MRI/CT scan capabilities. Carrying out compatibility testing, a crucial step in the development process, in order to make sure the platform works well with other emerging technologies and remains up to date with newer devices like the HoloLens 2.

All of these developments offer chances to improve the platform's functionality, make it easier to use, and raise its educational value for both inexperienced and seasoned clinicians.

6.3 Reflection

This project has instilled in me the lesson to keep believing, even when things don't seem to go as planned, and that the right intention is crucial for overcoming the tough phases of a project. I had never developed an application of this kind before, my supervisor's support encouraged me to take the leap, which filled me with hope. So far, Unity has

shown to be an amazing game engine, and I'm just getting started. I am excited to use this engine for a variety of developments.

I have no regrets about using Unity, so I would choose it again. My interest in expanding my knowledge and comprehension of modern technology began to grow when I first saw the HoloLens device, which I found to be fascinating.

6.4 Conclusion

Above all, this project has shown me that there are different hardware and tool options for creating mixed reality applications. Your patience and time are the most important investments. When you take on new challenges and see them through to the end, you develop a sense of respect for yourself.

Go out there and do work that inspire! ☺

REFERENCES

- Bai, H. et al. (2020) ‘A user study on mixed reality remote collaboration with eye gaze and hand gesture sharing’, Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems [Preprint]. doi:10.1145/3313831.3376550.
- Boehm, B.W. (1988) ‘A spiral model of software development and enhancement’, Computer, 21(5), pp. 61–72. doi:10.1109/2.59.
- Boehm, B.W. (1988) ‘A spiral model of software development and enhancement’, Computer, 21(5), pp. 61–72. doi:10.1109/2.59.
- Borycki, D. (2019) Programming for mixed reality with Windows 10, unity, VUFORIA, and Urhosharp. Pearson Education, Inc.
- Brun, H. et al. (2018) ‘Mixed reality holograms for heart surgery planning: First user experience in congenital heart disease’, European Heart Journal - Cardiovascular Imaging, 20(8), pp. 883–888. doi:10.1093/ehjci/jey184.
- Bulliard, J. et al. (2020) ‘Preliminary testing of an augmented reality headset as a dicom viewer during autopsy’, Forensic Imaging, 23, p. 200417. doi:10.1016/j.fri.2020.200417.
- Cartucho, J. et al. (2020) ‘Multimodal mixed reality visualisation for Intraoperative Surgical Guidance’, International Journal of Computer Assisted Radiology and Surgery, 15(5), pp. 819–826. doi:10.1007/s11548-020-02165-4.
- Chen, L. et al. (2017) ‘Recent developments and future challenges in medical mixed reality’, 2017 IEEE International Symposium on Mixed and Augmented Reality (ISMAR) [Preprint]. doi:10.1109/ismar.2017.29
- Chien, J.-C. et al. (2019) ‘Hololens-based AR system with a robust point set registration algorithm’, Sensors, 19(16), p. 3555. doi:10.3390/s19163555.
- Chien, J.-C. et al. (2019) ‘Hololens-based AR system with a robust point set registration algorithm’, Sensors, 19(16), p. 3555. doi:10.3390/s19163555.

Galati, R. et al. (2020) 'Experimental setup employed in the operating room based on virtual and mixed reality: Analysis of pros and cons in open abdomen surgery', *Journal of Healthcare Engineering*, 2020, pp. 1–11. doi:10.1155/2020/8851964.

Gsaxner, C. et al. (2023) 'The hololens in medicine: A systematic review and taxonomy', *Medical Image Analysis*, 85, p. 102757. doi:10.1016/j.media.2023.102757.

Herron, J. (2016) 'Augmented reality in medical education and training', *Journal of Electronic Resources in Medical Libraries*, 13(2), pp. 51–55. doi:10.1080/15424065.2016.1175987.

House, P.M. et al. (2020) 'Use of the mixed reality tool "VSI patient education" for more comprehensible and imaginable patient educations before epilepsy surgery and stereotactic implantation of DBS or stereo-EEG electrodes', *Epilepsy Research*, 159, p. 106247. doi:10.1016/j.eplepsyres.2019.106247.

Incekara, F. et al. (2018) 'Clinical feasibility of a wearable mixed-reality device in Neurosurgery', *World Neurosurgery*, 118. doi:10.1016/j.wneu.2018.06.208.

Jannin, P. and Collins, D.L. (2013) 'The state of the art of visualization in mixed reality image guided surgery', *Computerized Medical Imaging and Graphics*, 37(2), pp. 98–112. doi:10.1016/j.compmedimag.2013.01.009.

Keveleigh (2021) Camera setup in unity - mixed reality, *Mixed Reality | Microsoft Learn*. Available at: <https://learn.microsoft.com/en-us/windows/mixed-reality/develop/unity/camera-in-unity> (Accessed: 02 December 2023).

Kexugit (no date) Unity - developing your first game with unity and C#, *Unity - Developing Your First Game with Unity and C# | Microsoft Learn*. Available at: <https://learn.microsoft.com/en-us/archive/msdn-magazine/2014/august/unity-developing-your-first-game-with-unity-and-csharp> (Accessed: 03 December 2023).

Kumar, R.P. et al. (2020) 'Use of mixed reality for surgery planning: Assessment and Development Workflow', *Journal of Biomedical Informatics*, 112, p. 100077. doi:10.1016/j.yjbinox.2020.100077.

Kunz, B. (2019) Part 2: How to set up Unity's video player for 2D UI & 3D viewing, Heizenrader. Available at: <https://heizenrader.com/part-2-how-to-set-up-unitys-video-player-for-2d-ui-3d-viewing/> (Accessed: 02 December 2023).

Maasthi, M.J. et al. (2020) 'An interactive approach deployed for rhinoplasty using mixed reality', 2020 International Conference on COMMunication Systems & NETworkS (COMSNETS) [Preprint]. doi:10.1109/comsnets48256.2020.9027491.

Marco (2017) Spiral model for game development: Techniques to develop games, Gamedevelopertips. Available at: <https://gamedevelopertips.com/spiral-model-for-game-development/> (Accessed: 30 November 2023).

Nguyen, N.Q. et al. (2020) 'An augmented reality system characterization of Placement Accuracy in Neurosurgery', Journal of Clinical Neuroscience, 72, pp. 392–396. doi:10.1016/j.jocn.2019.12.014.

Nischita (2019) Playing videos on the Hololens Apps, Using Video Player component in Unity to play videos in your Hololens apps. Available at: <https://codeholo.com/2018/01/14/playing-videos-on-the-hololens-apps/> (Accessed: 02 December 2023).

Ong, S. (2017) Introduction to the HoloToolkit, SpringerLink. Available at: https://link.springer.com/chapter/10.1007/978-1-4842-2769-5_4 (Accessed: 03 December 2023).

Rositi, H. et al. (2020) 'Presentation of a mixed reality software with a HoloLens headset for a nutrition workshop', Multimedia Tools and Applications, 80(2), pp. 1945–1967. doi:10.1007/s11042-020-09687-8.

Sánchez-Margallo, J.A. et al. (2021) 'Application of mixed reality in medical training and surgical planning focused on minimally invasive surgery', Frontiers in Virtual Reality, 2. doi:10.3389/frvir.2021.692641.

Takagi, A. et al. (2000) 'Development of a stereo video see-through HMD for AR Systems', Proceedings IEEE and ACM International Symposium on Augmented Reality (ISAR 2000) [Preprint]. doi:10.1109/isar.2000.880925.

Technologies, U. (2023) Scenes, Unity. Available at: <https://docs.unity3d.com/Manual/CreatingScenes.html> (Accessed: 04 December 2023).

Technologies, U. (no date) Camera component, Unity. Available at: <https://docs.unity3d.com/Manual/class-Camera.html> (Accessed: 02 December 2023).

Tepper, O.M. et al. (2017) 'Mixed reality with hololens: Where virtual reality meets augmented reality in the operating room', *Plastic & Reconstructive Surgery*, 140(5), pp. 1066–1070. doi:10.1097/prs.0000000000003802.

Turini, G. et al. (2018) 'A Microsoft hololens mixed reality surgical simulator for patient-specific hip arthroplasty training', *Lecture Notes in Computer Science*, pp. 201–210. doi:10.1007/978-3-319-95282-6_15.

Uzun, Y. (2018) Real-Time Augmented Reality Collaboration with Different Device Types [Preprint]. doi:10.13140/RG.2.2.31908.01923.

Wake, N. et al. (2019) 'Patient-specific 3D printed and augmented reality kidney and prostate cancer models: Impact on patient education', *3D Printing in Medicine*, 5(1). doi:10.1186/s41205-019-0041-3.

Appendix A: Project Proposal for MSc in Computer Games Technology

Name: Shreyas Manjunath

E-Mail Address: shreyasmg88@gmail.com

Contact Phone number: 07735046813

Project Title: A Flexible and Modular Platform for presentation of Medical Data to Clinicians

Supervisor: Philip Edwards

Date: 10th December 2023

CITY, UNIVERSITY OF LONDON INM373 RESEARCH METHODS AND PROFESSIONAL ISSUES | 2022/23 | COURSEWORK | Shreyas Manjunath |220040817

Research proposal for INM363 MSc Individual Project (supervisor: Philip Edwards)

A Flexible and modular platform for presentation of medical data to clinicians.

Introduction

Head-mounted displays or HMDs are devices that are worn on the head that display images and information to the user. They are being used in the medical field to improve the way doctors and medical professionals learn and work. HMDs provide new and innovative methods for training medical professionals, diagnosing illnesses, and treating patients. Augmented reality (AR) and virtual reality (VR) are two technologies that have been suggested for use in surgical applications. Despite showing promise, few of these applications have been successfully put into standard clinical practice. A study by (Dilley et al., 2019) suggests that augmented reality is not ideal as it confuses the scene and that it is better to have the virtual picture close by rather than overlaid on the real view. The problem is that overlaid views on the patient don't really help that much as they obscure the normal view of the scene. However, a recent study from (Cartucho et al., 2020) looking at multiple visualizations concluded that a 3D model is the most popular mixed reality (MR) visualization component and mixed reality (MR) can be used in operating rooms for intraoperative surgical guidance.

This project will develop a flexible system for visualization of surgical and imaging data, where the location of each visualization can be placed and configured by the clinician. The aim is to see whether allowing the user to move and configure all of the visualizations at their fingertips will allow better decision making, training and planning of procedures. The proposed platform will combine various visualization and interaction models and it is hoped that allowing the user to have control will make it less confusing and thereby provide an improved user experience.

This flexible methodology will allow medical professionals and novices to use and interact with the visualization. Initially, the visualization will include video both 2D and stereo, rendering of the patient model which is a 3D view of anatomy and CT/MRI slice view. The application will be the discussion of finished surgical cases to mentor more junior doctors. The purpose is to instruct medical students and user in the proper positioning of visual elements, provide them with control over it, and allow them to determine how to utilize the application. The focus of the initial testing and evaluation will be on novices. Later on in the research, access to clinicians will be available.

Research Questions, Objectives, Beneficiaries

The following are the research questions underlying this project:

- Does 3D presentation of the video help in understanding the procedure?
- Does 3D rendering of the patient anatomy help in understanding the procedure?
- Is the ability to orient and configure the different visualizations helpful in conveying 3D relationship of structures?

- How can we involve this visualization platform in the healthcare setting to provide improved learning over traditional approaches?
- How can we use visualization components such as 3D objects and virtual 2D/3D screens to benefit clinicians and novices?

The research objectives to answer these questions are therefore:

- Research online clinical datasets for research;
- to research existing tools for medical visualization in Unity and to implement them;
- to view multiple visual components simultaneously using both 2D screens and 3D virtual objects;
- to implement an image viewer such as CT/MRI scans; and
- to investigate and involve interaction with novices and potentially clinicians to see what benefits they think each visualization provides.

The main intended beneficiaries of the research are medical industry professionals: surgeons, clinicians, nurses, medical students. I hope that this review will be helpful to clinicians and engineers working together on new mixed reality initiatives.

Motivation

I am a computer games technology student, working on a mixed reality project will challenge me to learn new technologies, tools, and techniques that can enhance my knowledge as a games student. It will allow me to explore different aspects of interactive technology, although this is not a games project I will be using interactive user components, developing in unity and using gaming devices like the HoloLens and Vive virtual reality (VR) Headset. By creating the proposed project for healthcare education, I will be contributing to improving medical training. This sense of purpose will immensely be fulfilling and motivating. Working on this project will involve collaboration with healthcare professionals this will expose me to different perspectives, challenges my assumptions, and fosters a spirit of teamwork. The proposed mixed reality project will push me outside of my comfort zone and encourages personal growth while improving my problem-solving skills, critical thinking abilities, and creativity. Undertaking a medical mixed reality project as a games student provides me with an opportunity for personal and professional growth, also enables me to make a real-world impact.

Key word: Mixed Reality, Rendering, Head-mounted-displays

Abbreviations: AR- Augmented reality, VR- Virtual reality, HMDs- Head-mounted displays

Critical Context

The project will mainly draw on concepts from medical image analysis, multimodal mixed reality visualization and use of the 3D visualization. The field of medical/surgical augmented reality (AR) is broad. There are several review papers that examine the study of visualization use in medical procedures. A recent study (Birlo et al., 2022) looks into optical see-through augmented reality, suggest that for augmented reality to be successful clinically, it is essential to mark the crucial points during particular surgical procedures where guidance is required. The paper from (Dilley et al., 2019) compares side by side imagine guidance and augmented

reality in intraoperative surgical guidance and says that using augmented reality enhanced the understanding of subsurface anatomy might improve the safety and effectiveness of surgical procedures, however, in terms of efficiency and safety, side-by-side picture guidance outperforms augmented reality.

(Cartucho et al., 2020) suggested that among surgeons, 3D model was found to be the most favoured visualization component and implies the use of mixed reality for enhancing surgical procedures. The study (Incekara et al., 2018) provides the proof of concept for the clinical feasibility of planning surgery in the operating room using the HoloLens device. According to the study by (Hyde et al., 2019) using interactive virtual 3D models has an impact on surgical choices and these models can change the surgical strategy and increase surgeon confidence.

These studies provide the use of visualization applications for surgical guidance or using the applications for performing intraoperative surgical guidance. In order to train both novices and medical professionals, the suggested system will be an mixed reality visualization application for teaching and debating surgical cases after they have been performed using both 2D/3D screens and 3D virtual objects. The proposed system will investigate the optimal combination of these visualizations for medical training.

Medical Image Processing

Medical image processing involves the use and study of image datasets of the human body. This project will make use of 3D images – from either a CT or an MRI scan to provide anatomical data to direct medical treatments like surgical planning or to be used as a teaching tool for novices. The investigation by (Birlo et al., 2022) examined the use of MRI/CT scans as a volumetric data source to help navigate during surgery. through 2D slices of volumetric data or 3D renderings. The conclusion of this review paper was that human factors need to be considered in the design of any guidance system, offering further support for the work proposed in this project. According to the study by (Melnyk et al., 2020) digital patient models are best preferred for basic anatomical and surgical training over human cadavers or real animals. The study by (Shin et al., 2016) showed that the 3D phantom models made it simple to see in 3D where the index lesion was inside the prostate and how big it was, the models assisted surgeons in determining how far or close the index cancer was to the prostate capsule. From the above studies there is a proof that the 3D models served as a helpful reference for surgeons both before and during surgery.

Once the data sets of the patient model is loaded the process of segmenting the images such that the required body part is generated and user is able to see the 3D model of the images. I will consider this tutorial by (Andijakl, 2020) to help me with the implementation. The analysis of (Bankman, 2009) says since measurements and other processing steps are dependent on segmented regions, segmentation is a separation stage that often gives a simple approach to the human visual system this technique of segmentation depicts the results of the complete study.

The type of visualization I aim to use in this project can be seen in the picture below. It shows a 3D model of the prostate next to an MRI of the same organ.

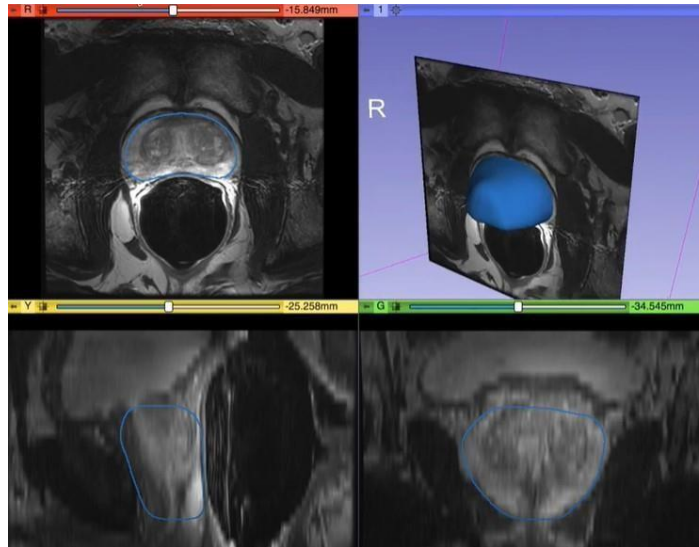


Figure 1 the above image shows the software deepinfer.org, these are the kind of slice and 3D views that will be implemented in this project. The image comes from the source: Deepinfer(2018a)

3D Visualization

Medical training and procedures are improved by the use of 3D visualization, which provides learners and practitioners with depth and clarity. (Hu et al., 2019) discuss different applications of mixed reality in medical field and how the use of mixed reality technology can provide an ideal platform for surgical training in the realistic 3D visualization world. There is a study by (Lin et al., 2019) which suggest that using 3D pictures early in a programme for teaching surgical residents can improve the anatomy-imaging-surgery system. The kind of visualizations that will be implemented in this project are shown in Figure 1.

It is clear that 3D visualization improves medical education and procedures but no study has been done to determine whether one visualization was superior to the other. Therefore, this project will create a questionnaire asking users and novices if a visualization was helpful or how it could be improved upon following use and seeking input from users and novices regarding what should be done in a certain case before perhaps organising a case.

Approaches: Methods & Tools for Design, Analysis & Evaluation

Methods and Tools for design:

The development environment for this project will be Visual Studio 2017, and the game engine will be Unity3D (Unity Technologies, San Francisco, CA, USA), version 2018.4.7f LTS. The HoloLens (1st Gen) has been utilized using the HoloToolkit 2017.4.23. The study by (Cartucho et al., 2020) suggested that the unity packages will be capable to be used on a variety of platforms, including a traditional 2D screen, stereoscopic anaglyph glasses, a dedicated 3D screen, a virtual reality headset for immersive experiences, or even a mixed reality headset that combines virtual and real-world elements.

Therefore, in unity platform the first part of the implementation will be comparison of 2D and 3D video, since there is already a video player in unity, 2D videos will be displayed using the platform's tools. To make the video appear to be played in 3 dimensions, I will play it twice once for each eye so that it will appear to be in 3D. I now have a unity implementation task to

complete. I will submit a questionnaire following this implementation to determine whether the 3D version of the video improves knowledge of the procedures. The platform will also implement a slice view of MRI or CT scans so that users can view particular cross-sectional scan slices, a virtual view is a digital representation or 3D model of the scan, a real-time video that can be directly overlaid with this virtual view. As mentioned before the tutorial by (Andijakl, 2020) will be used for the basis for this purpose of implementation.

The HoloLens/platform will have user experience (UX) building components, which will help in, for example, projecting virtual 3D models. Some of the interactions between the user and these components will involve moving, scaling, scrolling, or rotating the virtual objects in the virtual world. I will refer to (Polar-Key) mixed reality toolkit tutorials and documentations, which go over the fundamentals of the first-generation HoloLens. This tutorial will guide me in creating a basic mixed reality application using Unity.

Overall, this implementation aims to achieve that when the user places the HoloLens headset they will be able to see a virtual screen or a mixed reality screen and because of these interactions the user will be able to interact with and analyse the medical scans in a more simple and interactive way. Furthermore, the implementation of real-time video with the virtual view will be able to provide better accuracy and precision of medical training. I will hand out a questionnaire to users after they try the visualization to get their preferences for element placement and their opinions on how well it works.

A phantom, in the context of medical imaging, taken from an online source will be used as a 3D model for training and experimenting with the novices and users. When I have the right model, I will go ahead and incorporate it into the unity program. Besides improving the model's visualization, this interaction will also provide users with a more immersive experience. I will be using the Cancer Imaging library (TCIA), a site that de-identifies and hosts a sizable library of medical images of cancer that are available for public download, to obtain a suitable data set of images of patients with prostate cancer.

Implementation

To develop for HoloLens using unity, I need to install the necessary tools, including the universal windows platform (UWP) module, visual studio, and the windows software development kit (SDK). Once these are installed, create a new unity project and set the platform to UWP. Then, adjust the project settings to configure for HoloLens development. For 2D and 3D video implementation use the unity's video player component to play 2D video and use stereo video player for 3D video.

Implement a UI slider and/or voice command in HoloLens to enable users to navigate through different slices of a scan. To view MRI/CT scans virtually, they can be converted into 3D models using Slicer and then imported into Unity for holographic display. HoloLens spatial mapping and camera capabilities can be used to overlay the virtual view on a real-time video feed.

By importing the mixed reality Toolkit for unity, there will be access to essential scripts and prefabs for HoloLens development. The mixed reality toolkit's (MRTK) hand interaction prefabs provide ready-to-use solutions for common interactions like grabbing, rotating, and scaling, which can be customized to fit the needs of my application. The mixed reality toolkit's "Bounds Control" is essential for scaling, rotating, and moving objects, while the "Scrolling Object Collection" can be used for scrolling functionality. MRI/CT scans can be converted into 3D models (e.g., in OBJ or FBX format) and imported into unity, where they can be manipulated using mixed reality toolkit's interactions.

It's important for me to regularly test the application on the HoloLens device to make sure that the interactions are intuitive and the visualization is clear. Based on the results of these tests, I will make any necessary adjustments and when the results are satisfied, use unity's universal windows platform (UWP) build settings to build and deploy the application to HoloLens.

Analysis

In the early stages of implementation, there are a number of areas that require theoretical and practical study. As the combination of these tools offers a solid framework for creating this application, a thorough analysis of the functionality offered by the mixed reality toolkit must be conducted along with the capabilities of HoloLens, with the application of slicer and the use of unity. The study by (Maddali et al., 2022) provides a detailed analysis of the proposed interactions and interface, highlighting the significance of simple and effective interactions for medical visualization in a mixed reality setting. A 3D representation of an MRI or CT scan provides a comprehensive perspective of the region being scanned. This implementation will make use of a tutorial by Andijakl (2020), which implies that the implementation is depending on established methods or strategies.

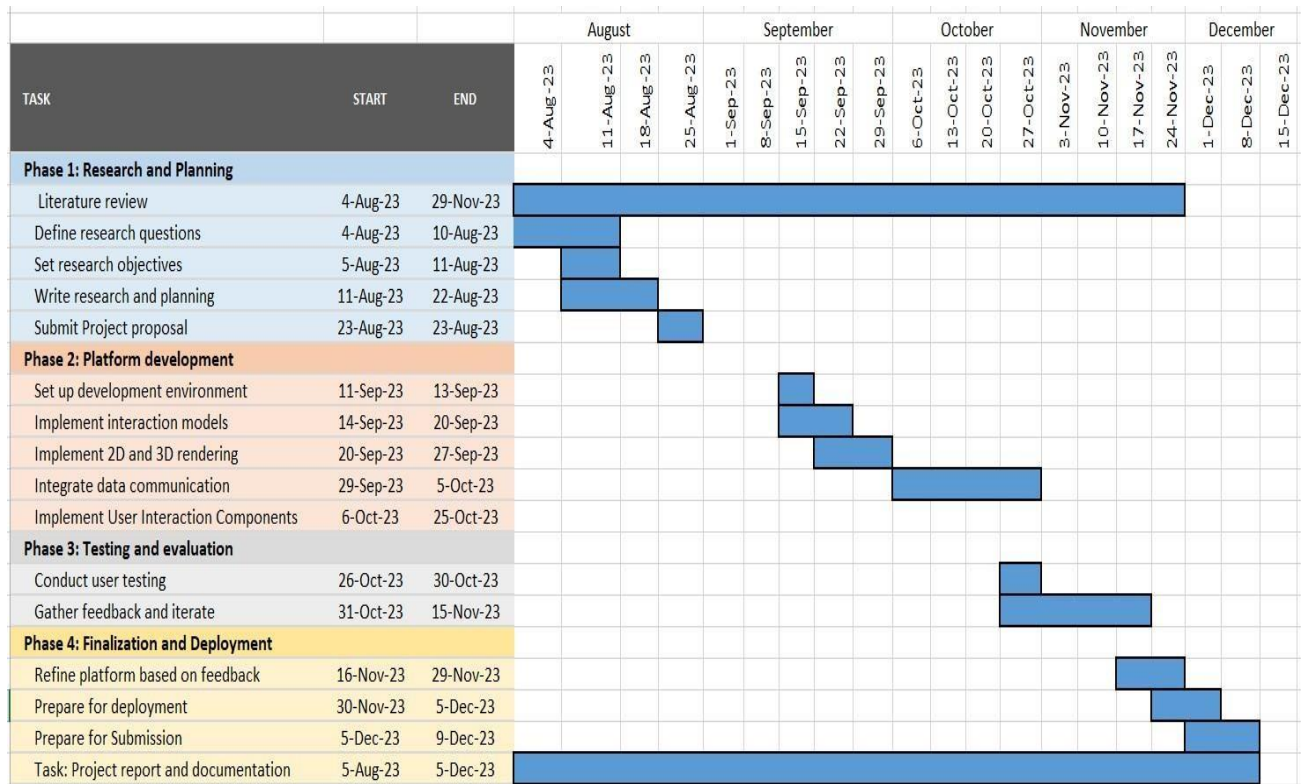
The accuracy and quality of the 3D model could be a challenge as I am relying on an online source for the phantom. For training to be effective, it is essential that the model should be accurate approximation of the human body. The aim is to create an immersive experience, but I also need to make sure that the model reacts to user input correctly and without any lags or glitches. Although the fact that the Cancer Imaging library (TCIA) de-identifies the medical images, data privacy and ensuring that no patient information can be linked back from the images can pose problems.

The platform will be made available to users and novices to engage with the visualization elements in order to assess the usability of the suggested visualizations, and following the interaction, a questionnaire will be made available. The aim of the questionnaire is to gather input on usability and the best ways to use the visualization components. Therefore besides the design and implementation, information overload could also be a major challenge of this project.

Evaluation

By collecting feedback through questionnaires and, possibly, by also capturing them while playing, evaluation based on user and novice experience will be carried out. If time permits, implement voice commands or gestures in HoloLens, so that users can respond to the questionnaire.

Work Plan



Comments and notes

The time allotted for each of the mentioned activities is shown by the bars in the graphical representation above. The project has been broken down into four distinct phases in order to ensure clarity and prevent confusion. The sequential start of each phase will allow me to concentrate on particular aims and objectives rather than juggling several tasks at once. This phased approach helps maintain a planned and organized workflow throughout the project and ensures a clear grasp of what needs to be done within each step.

This strategy makes sense because the interviews will need to be scheduled in advance and scheduled according to the availability of the interview subjects. On the other hand, it also calls for active project management discipline, careful planning, and constant awareness of dependencies and the critical path. Meetings with the project supervisor are not displayed because the times are not yet known.

Feasibility

Considering the limited timescale, creating a full platform that enables clinicians to easily see and interact with medical data while utilizing modern technology like HMDs and other visualisation approaches proves to be a difficult task.

I believe that with the efficient use of the suggested methods, implementation and resources the proposed project is feasible.

Risk Assessment

Work Phase	Risk	Likelihood of failure (L)	Impact of failure(I)	Severity (S = L x I)	Mitigation
Literature Review	Incomplete or outdated sources	1	6	6	Conduct a thorough and systematic literature search using relevant databases and keywords
Developing Environmental Setup	Technical issues or compatibility problems with the development environment, tools or hardware/software	3	7	21	Test and verify the functionality and compatibility of the development environment, tools and hardware/software before starting the development process
Interaction components Implementation	Ineffective interaction components for user input and interface interaction	3	7	21	High-rated tutorials and previously used components will serve as the foundation for implementation.
2D and 3D Rendering Development	Low quality or performance of the graphical components	3	10	30	Optimize the graphical components available in the Microsoft mixed reality tool kit development
Data set resources	Getting access to online archives	2	9	18	Request access to the archives as early as possible
Adding video player	Implementation in VR/MR proves difficult or there is lack of availability of the devices	3	10	30	Display on a screen as fall back
User Experience Component Implementation	Poor user experience by lack of graphical processing, rendering	2	7	14	Use low poly models for better user experience.

References

- Andijakl (2020) Visualizing MRI & CT Scans in mixed reality / VR / AR, part 1: Importing data, andreasjakl.com. Available at: <https://www.andreasjakl.com/visualising-mri-ct-scans-in-mixed-reality-vr-ar-part-1-importing-data/> (Accessed: 21 August 2023).
- Bankman, I.N. (2009) Handbook of Medical Image Processing and analysis. doi:10.1016/b978-0-12-373904-9.x0001-4.
- Birlo, M. et al. (2022) ‘Utility of optical see-through head mounted displays in augmented reality-assisted surgery: A systematic review’, *Medical Image Analysis*, 77, p. 102361. doi:10.1016/j.media.2022.102361.
- Cartucho, J. et al. (2020) ‘Multimodal mixed reality visualisation for Intraoperative Surgical Guidance’, *International Journal of Computer Assisted Radiology and Surgery*, 15(5), pp. 819–826. doi:10.1007/s11548-020-02165-4.
- Deepinfer (no date) DeepInfer. Available at: <http://www.deepinfer.org/> (Accessed: 22 August 2023).
- Deepinfer (2018a): Prostate segmenter, YouTube. Available at: <https://www.youtube.com/watch?v=BgqUzAwMfvQ> (Accessed: 22 August 2023).
- Dilley, J.W. et al. (2019) ‘Perfect registration leads to imperfect performance’, *Annals of Surgery*, 269(2), pp. 236–242. doi:10.1097/sla.0000000000002793.
- Hu, H. et al. (2019) ‘Application and prospect of mixed reality technology in medical field’, *Current Medical Science*, 39(1), pp. 1–6. doi:10.1007/s11596-019-1992-8.
- Hyde, E.R. et al. (2019) ‘Interactive virtual 3D models of renal cancer patient anatomies alter partial nephrectomy surgical planning decisions and increase surgeon confidence compared to volume-rendered images’, *International Journal of Computer Assisted Radiology and Surgery*, 14(4), pp. 723–732. doi:10.1007/s11548-019-01913-5.
- Incekara, F. et al. (2018) ‘Clinical feasibility of a wearable mixed-reality device in Neurosurgery’, *World Neurosurgery*, 118. doi:10.1016/j.wneu.2018.06.208.
- Lin, C. et al. (2019) ‘When to introduce three-dimensional visualization technology into surgical residency: A randomized controlled trial’, *Journal of Medical Systems*, 43(3). doi:10.1007/s10916-019-1157-0.
- Maddali, D. et al. (2022) ‘Spatial orientation in cardiac ultrasound images using mixed reality: Design and evaluation’, *Frontiers in Virtual Reality*, 3. doi:10.3389/frvir.2022.881338.
- Melnyk, R. et al. (2020) ‘Correction to: Mechanical and functional validation of a perfused, robot-assisted partial nephrectomy simulation platform using a combination of 3D printing and hydrogel casting’, *World Journal of Urology*, 38(7), pp. 1643–1643. doi:10.1007/s00345-020-03089-z.
- Polar-Kev (no date a) MRTK2-Unity Developer Documentation - MRTK 2, MRTK2-Unity Developer Documentation - MRTK 2 | Microsoft Learn. Available at: <https://learn.microsoft.com/en-us/windows/mixed-reality/mrtk-unity/mrtk2/?view=mrtkunity-2022-05#getting-started-with-mrtk> (Accessed: 22 August 2023).
- Shin, T., Ukimura, O. and Gill, I.S. (2016) ‘Three-dimensional printed model of prostate anatomy and targeted biopsy-proven index tumor to facilitate nerve-sparing prostatectomy’, *European Urology*, 69(2), pp. 377–379. doi:10.1016/j.eururo.2015.09.024.

Research Ethics Review Form: BSc, MSc and MA Projects

Computer Science Research Ethics Committee (CSREC)

<http://www.city.ac.uk/departments-computer-science/research-ethics>

Undergraduate and postgraduate students undertaking their final project in the Department of Computer Science are required to consider the ethics of their project work and to ensure that it complies with research ethics guidelines. In some cases, a project will need approval from an ethics committee before it can proceed. Usually, but not always, this will be because the student is involving other people ("participants") in the project.

In order to ensure that appropriate consideration is given to ethical issues, all students must complete this form and attach it to their project proposal document. There are two parts:

PART A: Ethics Checklist. All students must complete this part.

The checklist identifies whether the project requires ethical approval and, if so, where to apply for approval.

PART B: Ethics Proportionate Review Form. Students who have answered "no" to all questions in A1, A2 and A3 and "yes" to question 4 in A4 in the ethics checklist must complete this part. The project supervisor has delegated authority to provide approval in such cases that are considered to involve MINIMAL risk. The approval may be **provisional** – identifying the planned research as likely to involve MINIMAL RISK. In such cases you must additionally seek **full approval** from the supervisor as the project progresses and details are established. **Full approval** must be acquired in writing, before beginning the planned research.

A.1 If you answer YES to any of the questions in this block, you must apply to an appropriate external ethics committee for approval and log this approval as an External Application through Research Ethics Online - https://ethics.city.ac.uk/		<i>Delete as appropriate</i>
1.1	Does your research require approval from the National Research Ethics Service (NRES)? <i>e.g. because you are recruiting current NHS patients or staff?</i> <i>If you are unsure try - https://www.hra.nhs.uk/approvals-amendments/what-approvals-do-i-need/</i>	NO
1.2	Will you recruit participants who fall under the auspices of the Mental Capacity Act? <i>Such research needs to be approved by an external ethics committee such as NRES or the Social Care Research Ethics Committee - http://www.scie.org.uk/research/ethics-committee/</i>	NO
1.3	Will you recruit any participants who are currently under the auspices of the Criminal Justice System, for example, but not limited to, people on remand, prisoners and those on probation? <i>Such research needs to be authorised by the ethics approval system of the National Offender Management Service.</i>	NO
A.2 If you answer YES to any of the questions in this block, then unless you are applying to an external ethics committee, you must apply for approval from the Senate Research Ethics Committee (SREC) through Research Ethics Online - https://ethics.city.ac.uk/		<i>Delete as appropriate</i>
2.1	Does your research involve participants who are unable to give informed consent? <i>For example, but not limited to, people who may have a degree of learning disability or mental health problem, that means they are unable to make an informed decision on their own behalf.</i>	NO
2.2	Is there a risk that your research might lead to disclosures from participants concerning their involvement in illegal activities?	NO
2.3	Is there a risk that obscene and or illegal material may need to be accessed for your research study (including online content and other material)?	NO

2.4	Does your project involve participants disclosing information about special category or sensitive subjects? <i>For example, but not limited to: racial or ethnic origin; political opinions; religious beliefs; trade union membership; physical or mental health; sexual life; criminal offences and proceedings</i>	NO
2.5	Does your research involve you travelling to another country outside of the UK, where the Foreign & Commonwealth Office has issued a travel warning that affects the area in which you will study? <i>Please check the latest guidance from the FCO - http://www.fco.gov.uk/en/</i>	NO
2.6	Does your research involve invasive or intrusive procedures? <i>These may include, but are not limited to, electrical stimulation, heat, cold or bruising.</i>	NO
2.7	Does your research involve animals?	NO
2.8	Does your research involve the administration of drugs, placebos or other substances to study participants?	NO
A.3 If you answer YES to any of the questions in this block, then unless you are applying to an external ethics committee or the SREC, you must apply for approval from the Computer Science Research Ethics Committee (CSREC) through Research Ethics Online - https://ethics.city.ac.uk/ Depending on the level of risk associated with your application, it may be referred to the Senate Research Ethics Committee.		<i>Delete as appropriate</i>
3.1	Does your research involve participants who are under the age of 18?	NO
3.2	Does your research involve adults who are vulnerable because of their social, psychological or medical circumstances (vulnerable adults)? <i>This includes adults with cognitive and / or learning disabilities, adults with physical disabilities and older people.</i>	NO
3.3	Are participants recruited because they are staff or students of City, University of London? <i>For example, students studying on a particular course or module.</i> <i>If yes, then approval is also required from the Head of Department or Programme Director.</i>	NO
3.4	Does your research involve intentional deception of participants?	NO
3.5	Does your research involve participants taking part without their informed consent?	NO
3.5	Is the risk posed to participants greater than that in normal working life?	NO
3.7	Is the risk posed to you, the researcher(s), greater than that in normal working life?	NO
A.4 If you answer YES to the following question and your answers to all other questions in sections A1, A2 and A3 are NO, then your project is deemed to be of MINIMAL RISK. If this is the case, then you can apply for approval through your supervisor under PROPORTIONATE REVIEW. You do so by completing PART B of this form. If you have answered NO to all questions on this form, then your project does not require ethical approval. You should submit and retain this form as evidence of this.		<i>Delete as appropriate</i>
4	Does your project involve human participants or their identifiable personal data? <i>For example, as interviewees, respondents to a survey or participants in testing.</i>	YES

PART B: Ethics Proportionate Review Form

If you answered YES to question 4 and NO to all other questions in sections A1, A2 and A3 in PART A of this form, then you may use PART B of this form to submit an application for a proportionate ethics review of your project. Your project supervisor has delegated authority to review and approve this application under proportionate review. You must receive final approval from your supervisor in writing before beginning the planned research.

However, if you cannot provide all the required attachments (see B.3) with your project proposal (e.g. because you have not yet written the consent forms, interview schedules etc), the approval from your supervisor will be **provisional**. You **must** submit the missing items to your supervisor for approval prior to commencing these parts of your project. Once again, you must receive written confirmation from your supervisor that any provisional approval has been superseded by with **full approval** of the planned activity as detailed in the full documents. **Failure to follow this procedure and demonstrate that final approval has been achieved may result in you failing the project module.**

Your supervisor may ask you to submit a full ethics application through Research Ethics Online, for instance if they are unable to approve your application, if the level of risks associated with your project change, or if you need an approval letter from the CSREC for an external organisation.

B.1 The following questions must be answered fully. All grey instructions must be removed.		<i>Delete as appropriate</i>
1.1.	Will you ensure that participants taking part in your project are fully informed about the purpose of the research?	YES
1.2	Will you ensure that participants taking part in your project are fully informed about the procedures affecting them or affecting any information collected about them, including information about how the data will be used, to whom it will be disclosed, and how long it will be kept?	YES
1.3	When people agree to participate in your project, will it be made clear to them that they may withdraw (i.e. not participate) at any time without any penalty?	YES
1.4	Will consent be obtained from the participants in your project? Consent from participants will be necessary if you plan to involve them in your project or if you plan to use identifiable personal data from existing records. "Identifiable personal data" means data relating to a living person who might be identifiable if the record includes their name, username, student id, DNA, fingerprint, address, etc. <i>If YES, you must attach drafts of the participant information sheet(s) and consent form(s) that you will use in section B.3 or, in the case of an existing dataset, provide details of how consent has been obtained.</i> <i>You must also retain the completed forms for subsequent inspection.</i> <i>Failure to provide the completed consent request forms will result in withdrawal of any earlier ethical approval of your project.</i>	YES
1.5	Have you made arrangements to ensure that material and/or private information obtained from or about the participating individuals will remain confidential?	YES

B.2 If the answer to the following question (B2) is YES, you must provide details		<i>Delete as appropriate</i>
2	Will the research be conducted in the participant's home or other non-University location? <i>If YES, you must provide details of how your safety will be ensured.</i>	NO

B.3 Attachments ALL of the following documents MUST be provided to supervisors if applicable. All must be considered prior to final approval by supervisors. A written record of final approval must be provided and retained.	YES	NO	Not Applicable
Details on how safety will be assured in any non-University location, including risk assessment if required (see B2)			Not Applicable
Details of arrangements to ensure that material and/or private information obtained from or about the participating individuals will remain confidential (see B1.5) <i>Any personal data must be acquired, stored and made accessible in ways that are GDPR compliant.</i>			Not Applicable
Full protocol for any workshops or interviews**			Not Applicable
Participant information sheet(s)**	YES		
Consent form(s)**	YES		
Questionnaire(s)** <i>sharing a Qualtrics survey with your supervisor is recommended.</i>	YES		
Topic guide(s) for interviews and focus groups**			Not Applicable
Permission from external organisations or Head of Department** <i>e.g. for recruitment of participants</i>			Not Applicable

****If these items are not available at the time of submitting your project proposal, then *provisional***

approval can still be given, under the condition that you must submit the final versions of all items to your supervisor for approval at a later date. **All** such items **must** be seen and approved by your supervisor before the activity for which they are needed begins. Written evidence of **final approval** of your planned activity must be acquired from your supervisor before you commence.

Changes

If your plans change and any aspects of your research that are documented in the approval process change as a consequence, then any approval acquired is invalid. If issues addressed in Part A (the checklist) are affected, then you must complete the approval process again and establish the kind of approval that is required. If issues addressed in Part B are affected, then you must forward updated documentation to your supervisor and have received written confirmation of approval of the revised activity before proceeding.

Templates for Consent and Information

You must use the templates provided by the University as the basis for your participant information sheets and consent forms. You **must** adapt them according to the needs of your project before you submit them for consideration.

Participant Information Sheets, Consent Forms and Protocols must be consistent. Please ensure that this is the case prior to seeking approval. Failure to do so will slow down the approval process.

We strongly recommend using Qualtrics to produce digital information sheets and consent forms.

T +44 (0)20 7040 5060

Please tick
or
initial box

1.	I confirm that I have read and understood the participant information dated 06/12/2023 and Version 1.0 for the above study. I have had the opportunity to consider the information and ask questions which have been answered satisfactorily.	
2.	I understand that my participation is voluntary and that I am free to withdraw without giving a reason without being penalised or disadvantaged.	
3.	I agree to the focus group/interview being audio OR video recorded.	
4.	I agree to City recording and processing this information about me. I understand that this information will be used only for the purpose(s) explained in the participant information and my consent is conditional on City complying with its duties and obligations under the General Data Protection Regulation (GDPR).	
5.	I agree to take part in the above study.	

Name of Researcher [printed]	Signature	Date
------------------------------	-----------	------

B: 1



City, University of London
Northampton Square
London
EC1V 0HB
United Kingdom

T +44 (0)20 7040 5060

Participant Information Sheet

Title of study: A Flexible and modular platform for presentation of medical data to clinicians

Name of principal investigator: Shreyas Manjunath
Supervised by: Philip Edwards

Dear participant,

You are being invited to take part in a study to look at the effect of 3D perception on the understanding of anatomy and videos of a surgical scene. Before you decide whether you would like to take part it is important that you understand why the research is being done and what it would involve for you. Please take time to read the following information carefully and discuss it with others if you wish. Ask us if there is anything that is not clear or if you would like more information. You will be given a copy of this information sheet to keep.

What is the purpose of the study?

We are examining the effect of 3D perception and the ability to move objects using a mixed reality headset (the HoloLens 1 from Microsoft). This is like a virtual reality headset, but places virtual objects in the real world so you can see both together. We want to know if 3D rendering of a surgical scene as well as an anatomical model of the patient can improve education and understanding of the position of different structures. In particular we want to see if movement of the anatomical model and alignment to the scene can improve your understanding of what you see.

Why have I been invited?

You are being asked to take part in this study because you are an adult with good vision through both eyes with or without glasses, which can be worn while using the HoloLens device. Your participation is voluntary and you can withdraw at any time. You do not have to participate if you don't want to.

Do I have to take part?

Participation is voluntary, it is up to you to decide whether or not to take part. You can choose not to participate in part or all of the study. You can withdraw at any stage of the study without being penalised or disadvantaged in any way. If you do decide to take part you will be asked to sign a consent form. You are still free to withdraw at any time and without giving a reason.

www.city.ac.uk
and the professions

Academic excellence for business



City, University of London
Northampton Square
London
EC1V 0HB
United Kingdom

T +44 (0)20 7040 5060

What will happen if I take part?

You will be asked to wear the HoloLens device and to locate some objects in the room. These will consist of an image of a surgical scene in 2D and 3D and an anatomical model of the patient. These are from a kidney operation. You will be asked to place the anatomical model near to one of the images with correctly aligned orientation as you see it. You will then take a picture of the final alignment. This process is then repeated for the other image of the surgical scene. This process should take about 10 minutes.

What do I have to do?

After the interaction you will be asked to complete a short 10-15 minute survey about the experience.

What are the possible disadvantages and risks of taking part?

There are no foreseeable risks or harms or possible side effects for participating in this study. Your personal data, comments, and any other information you provide us will be de-identified or anonymised. We will keep all data confidential and secure.

What are the possible benefits of taking part?

While there are no specific benefits of taking part, we hope you enjoy the experience of participating in an observation. You may come away with a better understanding of how HoloLens is used in medical settings.

What will happen when the research study stops?

The findings from the observation, interviews, and materials will be anonymised and used in my dissertation report. It will only be shared with people involved in marking and moderating my dissertation.

Will my taking part in the study be kept confidential?

The data gathered in this study will be kept confidential. The saved images from the experiment will be kept on file for the purpose of this study only and no personal information will be stored.

What should I do if I want to take part?

Read, initial and sign the participant 'informed consent' form and remember to complete the task(s) as naturally as possible, even though you're being observed.

What will happen to the result of the research study?

The results of the observation will not be used for any purpose other than completing my dissertation.

What will happen if I do not want to carry on with the study?

www.city.ac.uk
and the professions

Academic excellence for business



City, University of London
Northampton Square
London
EC1V 0HB
United Kingdom

T +44 (0)20 7040 5060

You are free to withdraw from the study at any time without any explanation or penalty.

Who has reviewed the study?

Before any research is allowed to happen, it has to be checked by a Research Ethics Committee. This study has been approved by City, University of London – Computer Science Research Ethics Committee.

For further information about the research, or if you have any queries or concerns, please contact:

Shreyas Manjunath
Shreyas.Manjunath@city.ac.uk

Philip Edwards
Philip.Edwards@city.ac.uk
City, University of London
Northampton Square
London EC1V 0HB

Data Protection Privacy Notice: What are my rights under the data protection legislation?

City, University of London is the data controller for the personal data collected for this research project. Your personal data will be processed for the purposes outlined in this notice. The legal basis for processing your personal data will be that this research is a task in the public interest, that is City, University of London considers the lawful basis for processing personal data to fall under Article 6(1)(e) of GDPR (public task) as the processing of research participant data is necessary for learning and teaching purposes and all research with human participants by staff and students has to be scrutinised and approved by one of City's Research Ethics Committees.

The rights you have under the data protection legislation are listed below, but not all of the rights will apply to the personal data collected in each research project.

- right to be informed
- right of access
- right to rectification
- right to erasure
- right to restrict processing
- right to object to data processing
- right to data portability
- right to object
- rights in relation to automated decision making and profiling

www.city.ac.uk
and the professions

Academic excellence for business



City, University of London
Northampton Square
London
EC1V 0HB
United Kingdom

T +44 (0)20 7040 5060

For more information, please visit www.city.ac.uk/about/city-information/legal

What if I have concerns about how my personal data will be used after I have participated in the research?

In the first instance you should raise any concerns with the research team, but if you are dissatisfied with the response, you may contact the Information Compliance Team at dataprotection@city.ac.uk or phone 0207 040 4000, who will liaise with City's Data Protection Officer, Dr William Jordan, to answer your query. If you are dissatisfied with City's response you may also complain to the Information Commissioner's Office at www.ico.org.uk

What if there is a problem?

If the research is undertaken in the UK if you have any problems, concerns or questions about this study, you should ask to speak to a member of the research team. If you remain unhappy and wish to complain formally, you can do this through City's complaints procedure, which we will inform you of.

You could also write to the Secretary at:

Anna Ramberg
Research Integrity Manager – Research & Enterprise
City, University of London
Northampton Square, London, EC1V 0HB
Email: Anna.Ramberg.1@city.ac.uk

City holds insurance policies which apply to this study. If you feel you have been harmed or injured by taking part in this study you may be eligible to claim compensation. This does not affect your legal rights to seek compensation. If you are harmed due to someone's negligence, then you may have grounds for legal action.

Thank you for taking the time to read this information sheet and for taking part.

APPENDIX D: Participant Questionnaire

Section 1: Alignment and Interactions

Q1: how would you rate the ease of aligning the kidney model with the tumour in the application?

☐ 1 (Not easy)

☐ 2

☐ 3

☐ 4

☐ 5 (very easy)

Q2: How clear was the visualization of the kidney and the tumour in both the 3D and 2D video screens?

☐ 1 (Not clear)

☐ 2

☐ 3

☐ 4

☐ 5 (very clear)

Q3: Were you able to clearly see the blood vessels in the video? Were you able to understand the surgical scene better by aligning the model?

☐ 1 (not clear)

☐ 2

☐ 3

☐ 4

☐ 5 (clear)

Q4: Please rate the effectiveness of the 3D model in enhancing your understanding of the procedure:

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☐ 5 (very effective)

Q5: Please rate the effectiveness of using two hands to interact with the 3D model compared to using a single hand for the video screens interaction.

☐ 1 (less effective)

☐ 2

☐ 3

☐ 4

☐ 5 (very effective)

Q6: Which did you find more helpful for understanding the anatomy and the surgical procedure - the 3D video screen or the 2D video screen? Please explain your choice.

Reason:

Q7: Describe your experience interacting with the 3D anatomical patient model. What worked well, and what could be improved?

Description:

Section 2: Educational Value

Q8: How effective do you think this application is for educating users about surgical cases

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☐ 5 (effective)

Q9: Compared to traditional learning methods (like textbooks or lectures), how effective do you find this mixed reality application in providing educational content?

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☐ 5 (effective)

Q10: What improvements or additional features would you suggest to enhance the educational value of this application?

Suggestions:

Section 3: General Feedback

Q11: What was your overall experience using this application? Feel free to include any additional comments or suggestions.

Feedback:

Q12: In which of the following areas would you recommend using this platform?

☐ Practice Training

☐ Medical and Patient Education

☐ Surgery

☐ Rehabilitation

☐ Patient Treatment

APPENDIX E: Participant C - Questionnaire

Section 1: Alignment and Interactions

Q1: how would you rate the ease of aligning the kidney model with the tumor in the application?

☐ 1 (Not easy)

☐ 2

☐ 3

☒ 4

☐ 5 (very easy)

Q2: How clear was the visualization of the kidney and the tumor in both the 3D and 2D video screens?

☐ 1 (Not clear)

☐ 2

☐ 3

☐ 4

☒ 5 (very clear)

Q3: Were you able to clearly see the blood vessels in the video? Were you able to understand the surgical scene better by aligning the model?

☐ 1 (not clear)

☐ 2

☐ 3

☒ 4

☐ 5 (clear)

Q4: Please rate the effectiveness of the 3D model in enhancing your understanding of the procedure:

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☐ 5 (very effective)

Q5: Please rate the effectiveness of using two hands to interact with the 3D model compared to using a single hand for the video screens interaction.

☐ 1 (less effective)

☐ 2

☐ 3

☐ 4

☒ 5 (very effective)

Q6: Which did you find more helpful for understanding the anatomy and the surgical procedure - the 3D video screen or the 2D video screen? Please explain your choice.

Reason: 3D video screen is more helpful due to the visualisation of anatomy with appropriate depth of field.

Q7: Describe your experience interacting with the 3D anatomical patient model. What worked well, and what could be improved?

Description: Interacting with the 3D anatomical model was easier due to the degree of freedom of anatomical model across multiple axis though the movement of model in the up-down axis wasn't smooth compared to other axis, which can be improved.

Section 2: Educational Value

Q8: How effective do you think this application is for educating users about surgical cases

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☒ 5 (effective)

Q9: Compared to traditional learning methods (like textbooks or lectures), how effective do you find this mixed reality application in providing educational content?

☐ 1 (not effective)

☐ 2

☐ 3

☐ 4

☒ 5 (effective)

Q10: What improvements or additional features would you suggest to enhance the educational value of this application?

Suggestions: Providing well defined instruction to use this application or a user instruction video would reduce the time to understand the controls of virtual model. Providing a coordinate system/icon would provide the real time axis information of the model.

Section 3: General Feedback

Q11: What was your overall experience using this application? Feel free to include any additional comments or suggestions.

Feedback: Overall experience was good. As I learned this test involved HoloLens version 1.0, I reckon next generation models would be more precise and accurate which would be more suitable to be involved in the field of medical. Though the version 1.0 is great for educational purposes. As this technology evolve to higher accuracy, HoloLens has the potential to conduct surgeries in association with robotics.

Q12: In which of the following areas would you recommend using this platform?

☐ Practice Training

☒ Medical and Patient Education

☐ Surgery

☐ Rehabilitation

☐ Patient Treatment